Natural Origin V – The Space Unit Model of Light-Speed Refresh and Ontological Rewriting of Physical Laws

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Abstract

This paper proposes and systematically constructs a unified physical framework in which the space unit SU is the sole physical entity. The space unit has a discrete refresh structure, with a constant refresh relay speed of light (c), and within each refresh cycle, only two functionally orthogonal phase directions are allowed: the "stepping phase" (responsible for migration and propagation) and the "settling phase" (responsible for return and existence). All physical phenomena are reinterpreted as readings of the refresh budget distributed between these two orthogonal phases, subject to the geometric and topological conditions of phase closure.

In this framework, the invariance of the speed of light is no longer an empirical axiom or a result of spacetime symmetry, but a hardware-level invariance of the space unit's refresh protocol. Speed, time, energy, mass, and momentum are all strictly defined as readings of the refresh budget distribution; relativistic effects are thus unified as changes in these readings, rather than deformations of spacetime. Lorentz transformations are proven to be the reading transformations that preserve the decomposition of the refresh budget, and the equivalence principle arises from the spatial distribution of the settling-phase proportion.

The paper further rewrites the ontological structure of electromagnetism. Charge is defined as the minimal phase-step placeholder bias, rather than a material property; magnetic fields are strictly characterized as the directional closed structure forced by the settling-phase under the migration of the stepping phase; Lorentz force is an orthogonal geometric correction that arises to access this closed channel. Through discrete SU grids and auditable derivations, this paper provides a unified source for Maxwell's equations, proving them as reading-layer equivalence constraints for phase-closure legality, rather than empirical axioms.

On the microscopic level, the paper proves that the electron is the minimal stable enclosure state allowed within this framework, and its right-handed double-helix structure naturally leads to spin 1/2, magnetic moments, and non-radiative properties. Protons and neutrons are characterized as the second stable enclosure level, composed of three orthogonal double-helix pairs, and their mass scales and stability arise from enclosure-level transitions, not independent fundamental interactions. Atomic energy levels and orbits are strictly derived as integer spectra of pure stepping-phase closure, with angular momentum quantization and the $1/n^2$ energy level rule being direct results of closure integration conditions.

On the quantum level, this paper introduces the ontological distinction between "propagation state" and "settling state," providing a unified explanation for wave-particle duality, the measurement problem, and the origin of uncertainty. Entanglement is reinterpreted as multi-interface readings of an unresolved propagation ledger. By fully replicating experimental processes (preparation, propagation, detection, and counting) and deriving the CHSH inequality mathematically, this paper proves that the violation of Bell inequalities arises from the failure of

the separability factorization assumption in unresolved states, while strictly satisfying the no-signaling condition.
On the macroscopic level, the paper rewrites Newton’s laws, proving them as the macroscopic limit of "closed without residue." The sun’s rotation is revealed as a necessary geometric solution for closed stepping phases within stars, while planetary orbits arise from the external closure condition in the settling-phase anchor field. Newtonian gravity, Kepler's laws, and perihelion precession naturally emerge in the weak-field limit. Black holes are defined as dual-criterion limit states, where the settling phase is exhausted and the stepping closure scale reaches its minimum limit. The paper provides a strict proof of the "density consistency theorem" with no singularities, offering a new ontological foundation for black hole thermodynamics and the area law.
In the NO5 (Natural Origin V) framework, the differences between microscopic and macroscopic phenomena do not arise from different physical laws, but from different collaboration modes of stepping and settling phases during space-unit refresh. When the stepping phase dominates and settling is delayed, the incomplete closure exists as a propagation state, manifested in microscopic phenomena like superposition, interference, and probabilistic readings; when the settling phase continues to dominate and forces residuals to self-consume, the stepping phase is tamed by closure conditions, and the system behaves as a stable structure with deterministic trajectories, as in classical mechanics. Therefore, quantum and classical are not two separate physical systems, but two reading representations of the same refresh structure under different phase collaboration modes.
Finally, the paper discusses the engineering prospects of settling-phase protocols, pointing out that mass reduction and space propulsion are not anti-gravity or thrust technologies, but controllable rewrites of refresh-loop throughput capacity and settlement-anchor gradients, and outlines the future of field communication and continuous computers based on synchronization of settlements.
This paper provides a self-consistent physical framework that spans from the microscopic to the cosmic scale, and lists multiple auditable experimental and observational criteria.

Introduction

Modern physics has achieved unprecedented success over the past century, but its foundational structure has exhibited noticeable layered fragmentation. Quantum mechanics, classical electromagnetism, relativity, and gravitational theory are highly precise within their respective domains of applicability, yet they remain difficult to unify on the ontological level: issues such as the dual language of particles and fields, the physical status of wavefunctions, the special nature of measurement collapse, the origin of the invariance of the speed of light, and black hole singularities still rely on effective theories patched together to avoid resolution.
A common characteristic that runs through these problems is that existing theories predominantly start from the reading layer—energy, force, field, probability, and spacetime geometry—without addressing a deeper question: why are these readings possible, stable, and share common limit structures? As a result, questions like why charge is quantized, why magnetic fields must appear, why inertia exists, why orbits close, and why entanglement is correlated but uncommunicative can only be treated within different theories, lacking a consistent root across scales.
This paper takes a different starting point. Instead of treating particles or continuous fields as fundamental entities, we assume that the natural world consists of space units with discrete refresh

structures. Each space unit operates with a fixed refresh budget, and its relay speed is always the speed of light (c). More crucially, the refresh budget allows distribution between two orthogonal phase directions in each refresh cycle: the stepping phase (responsible for migration and propagation) and the settling phase (responsible for return and existence). All observable physical quantities can be understood as readings after the refresh budget is allocated between these two phases, and the geometric and topological results of whether phase closure can be completed.

This starting point immediately provides an ontological explanation for the invariance of the speed of light. The speed of light is no longer an empirical limit or a result of spacetime symmetry, but an intrinsic invariance of the space unit's refresh protocol: once a refresh relay fully shifts into the stepping phase, its inherent speed must be equal to c. From this perspective, relativistic effects no longer stem from the stretching or bending of spacetime, but from the redistribution of the refresh budget between the stepping and settling phases, leading to changes in readings.

The goal of this paper is not to introduce new correction terms or parameters, but to provide a self-consistent, auditable, and falsifiable logical chain showing that, starting from the refresh structure of space units, one can unify and derive the core structures of electromagnetism, quantum phenomena, relativity, classical mechanics, and black hole limits without introducing additional ontological assumptions. To avoid conceptual confusion, this paper always distinguishes between the "ontological layer" (refresh structure and closure conditions) and the "reading layer" (force, field, energy, time, etc.), and clearly specifies the mapping between them in each chapter.

The structure of this paper follows a unidirectional logic from ontological foundations to reading layers, from microscopic to cosmic scales, and from theory to engineering prospects. Chapters 1 and 2 establish the axiomatic structure of space units, orthogonal phases, and light-speed refresh, and provide the unified interface for all physical readings; subsequent chapters then rewrite electromagnetism, microscopic enclosure states, atomic structure, quantum entanglement, relativity, classical mechanics, black holes, and engineering limits, and finally discuss the prospects for mass reduction, space propulsion, field communication, and continuous computers. Appendices provide the physical legitimacy of calculus and continuous limits within this framework.

We emphasize that this paper does not aim to deny the success of existing theories within their domains of applicability, but to provide a common structural root for these successes. If these derivations are correct, the unification of physics no longer depends on higher energies or more complex mathematical forms, but on a deeper rethinking of "why existence is possible."

Chapter 1: The Ontological Axiom of Space Units, Orthogonal Phases, and Light-Speed Refresh

Goal of This Chapter

The goal of this chapter is not to explain specific physical phenomena, but to establish the minimal ontological axioms of the entire theory. All subsequent chapters—whether on charge, magnetic fields, quantum transitions, relativity, black holes, or future engineering—will only use the structure provided in this chapter, without introducing additional assumptions.

This chapter answers three key questions:

1. What is the minimal physical entity in nature?

2. What is the minimal degree of freedom structure allowed within this entity?
3. Why does the speed of light remain invariant in all physical processes?

1.1 Space Units (SU): The Sole Physical Entity

Definition 1.1 (Space Unit)

The minimal physical entity in nature is not particles, fields, or continuous spacetime, but the space unit SU, which has a discrete refresh structure.

Each SU:

- Is not a "point" but the smallest realizable refresh unit;
- Does not carry "material properties" in the traditional sense;
- Does not assume force, field, mass, or time beforehand.

The only meaning of existence is: an SU participates in refreshing.

1.2 Light-Speed Refresh: Hardware-Level Invariance of SU

Axiom 1 (Light-Speed Refresh Axiom)

The refresh relay speed of each SU is constant and equal to the speed of light (c), and this speed is independent of:

- the reference frame;
- the state of the SU;
- the surrounding environment or scale.

This axiom is not an empirical law, but a hardware constraint of the refresh protocol: refresh is a minimal time-space relay operation, and the ratio of the shortest completion time to the minimal spatial step is fixed at c. Therefore, the speed of light is not a "signal speed" or "propagation limit" but the intrinsic speed of the refresh itself.

1.3 Orthogonal Phases: The Minimal Decomposition of the Refresh Budget

Axiom 2 (Orthogonal Phases Axiom)

The refresh budget of an SU in any refresh cycle is only allowed to decompose into two, and only two, orthogonal phase directions:

1. Stepping Phase (ϕ_s)
- Describes the migration and execution of refresh in space
- Responsible for "motion," "propagation," and "acceleration"
- Cannot independently complete closure.

2. Settling Phase (ϕ_r)
- Describes the return and locking of refresh in the local structure
- Responsible for "existence," "mass," and "time readings"
- Is the only phase direction that can close.

This is not a coordinate choice but a functional orthogonality.

1.4 The Two Degrees of Freedom of Space Units, Temporal Rhythms, and the First-Principles Definitions of Energy, Mass, and Momentum

In the NO5 (Natural Origin V) framework, all physical quantities must be traced back to the available refresh budget within a single refresh cycle of space units (SU).

This section, without introducing any additional degrees of freedom or altering existing symbol

conventions, provides the degree-of-freedom structure of the space unit and the first-principles definitions of time, energy, mass, and momentum.

1.4.1 The Two Orthogonal Refresh Degrees of Freedom of Space Units

Space units (SU) are not point-like geometric objects, but the smallest physical execution units that refresh at a fixed intrinsic principal frequency.

In any given refresh cycle, the refresh budget of an SU can only be allocated between two, and only two, orthogonal degrees of freedom:

Stepping Degree (Step, degree s), with refresh component Ω_s

Settling Degree (Relay, degree r), with refresh component Ω_r

These components satisfy a strict orthogonality and conservation relationship:

$$\Omega_0^2 = \Omega_s^2 + \Omega_r^2$$

where:

- Ω_0: the intrinsic refresh principal frequency of the space unit (hardware-level upper limit)
- Ω_s: the refresh component used for executing the stepping phase
- Ω_r: the refresh component used to complete the settling and locking phase

This relationship is not a dynamical assumption but a computational bookkeeping constraint at the hardware level that space units must satisfy.

1.4.2 Clarification of the Physical Meaning of Stepping (s) and Settling (r)

1) Stepping Degree Ω_s: The phase execution demand

Stepping corresponds to the phase execution process that has not yet completed settling. Its physical manifestation includes:

- motion and velocity
- momentum
- propagation and current
- the objects affected by Lorentz force and other geometric corrections

In this paper, the stepping degree (s) describes the phase execution demand that has not yet completed self-cancellation. It does not form an independently existing or independently propagating phase entity but is an intermediate phase before settlement completion.

Therefore, we summarize: Stepping Degree (s) = Proportional Refresh of Phase Execution Demand

2) Settling Degree Ω_r: The phase return and locking

Settling corresponds to the phase returning and locking in the allowed closure channels. Its physical manifestation includes:

- position
- mass
- static existence
- gravitational source terms

Any structure that can exist stably and be read as an "entity" must have $\Omega_r \neq 0$. Therefore, we summarize: Settling Degree (r) = Proportional Refresh of Phase Closure and Locking

1.4.3 First-Principles Definition of Time (Not a Degree of Freedom)

In NO5, time is not a third spatial degree of freedom, nor is it an independent quantity attached to the stepping or settling axes.

The definition of time is: the rhythm count experienced by a space unit in the right-handed closed-loop refresh between the stepping (s) and settling (r) orthogonal degrees of freedom.

Thus:

- Time is not Ω_s
- Time is not Ω_r
- Time is the equal-weight rhythm formed by both under orthogonal conditions

Here, "equal-weight" simply means that execution and settlement participate equally in the count, not corresponding to any spatial or spacetime direction and introducing no new geometric dimensions.

This definition immediately gives:

- time dilation
- intrinsic clock slowing
- changes in time readings in relativity

Without introducing additional dimensions or assuming independent time flow.

1.4.4 First-Principles Definition of Energy: Total Refresh Strength

In this paper, the term "energy" has different meanings in the ontological layer and the reading layer: in the ontological layer, it refers to the upper limit of refresh strength; in the reading layer, it refers to the counting of settling readings.

Energy is not an entity stored in objects but a scalar reading of the available refresh strength of the space unit.

Definition: $E \equiv \hbar\,\Omega_0$

Its physical meaning is:

- Energy represents the total execution ability of a space unit within a single refresh cycle.
- $\hbar$ is the minimal action scale corresponding to a single refresh.

For a specific physical system, Ω_0 represents the effective refresh principal frequency the system can call upon; the change in the system's energy reflects changes in the distribution of Ω_s and Ω_r, rather than a change in the hardware limit.

1.4.5 First-Principles Definitions of Mass and Momentum through Refresh Accounting

By the refresh accounting relationship, we can directly obtain the first-principles definitions of mass and momentum:

Mass (settling component) $mc^2 = \hbar\,\Omega_r$

Its meaning is:

- Mass represents the portion of the refresh budget that is allocated to settling (r).
- The larger the mass, the higher the proportion of settling, and the more "resistant" the structure is to being dragged into pure propagation state.

Momentum (stepping component) $pc = \hbar\,\Omega_s$

Its meaning is:

- Momentum represents the portion of the refresh budget allocated to executing the stepping phase (s).
- All motion, inertia, and geometric mechanical corrections arise from Ω_s.

These definitions are scalar results of refresh accounting in the reading layer and do not require direct measurement within a single refresh cycle.

1.4.6 Physical Completion of the Refresh Pythagorean Relationship

By substituting the definitions of mass and momentum into the refresh conservation equation, we immediately obtain: $E^2 = (mc^2)^2 + (pc)^2$

This relationship is not derived from relativistic assumptions, but is a geometric identity of space

unit refresh accounting. The energy-momentum relationship in relativity appears in this paper as the manifestation of this identity in the continuous reading limit.

1.4.7 Chapter Summary

Space units have two orthogonal refresh degrees of freedom:

- Stepping (s) is responsible for phase execution, motion, and propagation;
- Settling (r) is responsible for phase closure, locking, and existence.

Time is not a degree of freedom but the rhythm count formed by the two under the right-handed closed-loop refresh.

Energy is the total refresh strength, and mass and momentum are the projections of refresh in the settling and stepping directions.

Thus, the core relationships of classical mechanics, relativity, and quantum mechanics are unified at the hardware level of space unit refresh, without introducing any additional degrees of freedom or ontological assumptions.

1.5 Ontological Explanation of the Invariance of the Speed of Light (Core)

In NO5, the invariance of the speed of light is not a result of spacetime symmetry but of a deeper fact: once a refresh relay fully shifts into the stepping phase, its inherent speed must be equal to c.

Thus:

- Light (propagation state) must propagate at c;
- Any enclosure state that loses settling closure must also propagate at c;
- No refresh relay can be “faster” or “slower.”

This explains:

- the invariance of the speed of light in a vacuum;
- the independence of the speed of light from the source or observer;
- the appearance of the speed of light as an unbreakable upper limit in all theories.

It is not an “assumption,” but a direct reading from the refresh protocol.

1.6 Closure and Existence: Physical Structure Filtering Principles

Axiom 3 (Closure and Existence Axiom)

Any physical structure that can exist stably and be observed as an "entity" must satisfy: in each refresh cycle, the residuals generated by the stepping phase must be completely absorbed by the settling phase.

Otherwise:

- If residuals are expelled → propagation state (radiation, light);
- If residuals accumulate → structure instability, disintegration.

Thus, existence is not a default; existence is filtered through closure conditions.

1.7 Clarification of the Role of Observers and Reference Frames

In NO5:

- No “absolute space” needs to be introduced;
- Nor does the “observer” need to be elevated to an ontological entity.

All descriptions are based on one convention: For an object at rest relative to some observer, the surrounding space's refresh structure can be clearly described.

This convention is used only for narration and does not affect the physical ontology.

1.8 Equivalence Statement (Unified Position of the Series of Papers)

In earlier works in this series, we repeatedly used the description: "For an object at rest relative to an observer, its surrounding space undergoes helical flow at the speed of light in cylindrical form." In the language of this chapter, this statement can be strictly rewritten as: "The space surrounding this object undergoes a refresh relay at the speed of light, and its refresh budget is distributed between stepping and settling phases, forming an axisymmetric helical structure under closure constraints."

The equivalence conclusion "space undergoes helical flow at the speed of light" and "space units refresh at light speed" describe the same physical fact in two equivalent forms. The former is a geometric-intuitive description, and the latter is an ontological structural description.

1.9 Chapter Summary

1. The minimal ontological entity in nature is the space unit (SU), not particles or fields;
2. The refresh relay speed of each SU is constant at light speed (c);
3. The refresh budget can only decompose into two orthogonal phase directions: stepping phase and settling phase;
4. The invariance of the speed of light arises from the refresh protocol itself, not from spacetime assumptions;
5. Existence is filtered through phase closure conditions;
6. The descriptions of "space undergoes helical flow at the speed of light" and "space units refresh at light speed" are physically equivalent.

Starting from this chapter, the speed of light is no longer an empirical limit, but the starting point of the natural world's refresh structure.

Chapter 2 Readouts, Relativity, and the Continuous Limit: From Space-Unit Calibration to a Unified Interface for Physical Quantities

Goal of This Chapter

This chapter accomplishes three tasks:

1. to state the physical origin and calibration of the space unit (SU), obtained from the inverse calibration in Natural Origin I using the harmonic structure of fine spectra and the fine-structure constant (α);
2. to provide unified definitions for all "readout quantities": velocity, time, energy, mass, and momentum;
3. to clarify the origin of the continuous limit and relativistic effects: they are not spacetime deformations, but readouts of how the refresh budget is apportioned between two phases.

2.1 Origin and Calibration of the Space Unit (Required)

2.1.1 Origin: Inverse Calibration from Fine-Spectral Harmonics and α

The space unit is not introduced as an ad hoc assumption. It arises from the spectroscopic inverse calibration developed in Natural Origin I:

- the harmonic (frequency-multiplicity) structure of the hydrogen fine spectrum is taken as the experimental anchor;
- the fine-structure constant α is used as the dimensionless constraint;

- observable spectral harmonics are mapped to refresh–enclosure hierarchy, and the minimal realizable refresh scale is obtained by inversion.

This procedure does not rely on unknown constants; it is a reverse construction from experiment → structure → ontology.

2.1.2 Numerical Values of the Space Unit

The calibration yields the SU scale and intrinsic refresh frequency:

$$\ell_0 \approx 3.76 \times 10^{-47}\, m \tag{2.1}$$

$$\Omega_0 \equiv \frac{c}{\ell_0} \approx 7.97 \times 10^{54}\, s^{-1} \tag{2.2}$$

Interpretation: Ω_0 is the intrinsic angular frequency of refresh relay, not a directly observable oscillation frequency; it is the counting baseline underlying all observable frequencies. Fine spectra are not the ontological origin of ℓ_0, but its only readable projection at atomic scale; the physical status of ℓ_0 is fixed by the SU refresh protocol itself.

2.2 Refresh Budget and Phase Fractions (Master Readout Identity)

Following Chapter 1, let the total refresh budget of an SU be Ω_0, decomposed into two orthogonal phases: $\Omega_0^2 = \Omega_s^2 + \Omega_r^2$

Define the dimensionless fractions: $\beta_s = \frac{\Omega_s}{\Omega_0}, \quad \beta_r = \frac{\Omega_r}{\Omega_0}$

Then the central identity of the entire book is:

$$\beta_s^2 + \beta_r^2 = 1 \tag{2.3}$$

This is not a dynamical equation, but a geometric decomposition of the refresh budget.

2.3 Readout Definitions of Velocity and Time (Root of Relativity)

2.3.1 Velocity Readout

When the refresh budget is allocated fractionally to the stepping phase, the readout of linear motion is:

$$v = \beta_s\, c \tag{2.4}$$

The speed of light (c) never changes; what changes is the fraction assigned to stepping.

2.3.2 Time Readout

Time is not an external parameter, but the counting density of settlement events. Thus:

$$d\tau = \beta_r\, dt \tag{2.5}$$

When β_r decreases, settlement events become sparser and the time readout slows accordingly.

2.4 Ontological Origin of Relativistic Effects (Without Spacetime Deformation)

From (2.3)–(2.5) we obtain directly:

- Lorentz factor:

$$\gamma = \frac{1}{\beta_r} \tag{2.6}$$

- time dilation: $d\tau = \frac{dt}{\gamma}$

- length contraction (readout interpretation): not that space itself is compressed, but that the settlement-phase budget is occupied by the stepping phase, reducing the number of spatial addresses that can be registered per unit time. In NO5, relativity is a readout effect, not a physical deformation of spacetime.

2.5 Unified Readouts of Energy, Mass, and Momentum

In this paper, "energy" has different meanings in the ontological layer and the readout layer: the former refers to the refresh upper bound, while the latter refers to the valuation of settlement readouts.

2.5.1 Energy

Energy is the readout amplified by settlement of the refresh budget:

$$E = \gamma\, m_0\, c^2 \tag{2.7}$$

Here m_0 is the settlement-scale readout of the enclosure structure at $\beta_s = 0$.

2.5.2 Mass

Mass is not "amount of substance," but the settlement-budget scale required to maintain existence. The effective mass readout is amplified as β_r changes: $m_{eff} = \gamma m_0$

2.5.3 Momentum

Momentum is the linear readout of the stepping phase:

$$\mathbf{p} = \gamma m_0 \mathbf{v} \tag{2.8}$$

2.6 Why the Continuous Limit Holds (Physical Root of Derivatives)

In NO5, the primitive operation is differencing; continuous derivatives arise as statistical limits.

- minimal time step: $\Delta t_0 = \ell_0 / c$
- any macroscopic derivative is defined by:

$$\frac{df}{dt} \equiv \lim_{N\to\infty} \frac{1}{N\Delta t_0} \sum_{k=1}^{N} f(t + k\Delta t_0) - f(t + (k-1)\Delta t_0)$$

The physical condition for the existence of this limit is that observable quantities must be insensitive to single-refresh noise; otherwise they are not repeatable and not measurable. This provides the ontological justification for using calculus in physics.

2.7 Failure Boundary of the Continuous Limit (Important Preview)

Continuous descriptions are not valid everywhere. When the observation scale satisfies $\lesssim \max(\ell_0, c/\omega_{\max})$ or when the settlement fraction approaches the limit $\beta_r \to 0$ (black holes, strong engineering working points), the continuous limit fails and one must return to the discrete SU description.

This criterion will be used in:

- Chapter 10 (orbits and precession);
- Chapter 13 (black holes and density consistency);
- Chapter 15 (mass reduction and propulsion).

2.8 Interface Role of This Chapter in the Whole Book

- Chapter 1: ontological axioms (SU, two phases, light-speed refresh);

- Chapter 2: readout and limit interface (relativity, energy, time, continuity);
- from Chapter 3 onward: all concrete physical and engineering manifestations.

After this chapter, the rest of the paper requires no additional "relativity assumptions."

2.9 Chapter Summary

1. The space unit (SU) is obtained by inverse calibration from fine-spectral harmonics and the fine-structure constant;
2. $\ell_0 \approx 3.76 \times 10^{-47}\, m,\ \Omega_0 \approx 7.97 \times 10^{54}\, s^{-1}$;
3. the refresh budget is apportioned between stepping and settlement phases, satisfying $\beta_s^2 + \beta_r^2 = 1$;
4. velocity, time, energy, mass, and momentum are all readouts of this apportionment;
5. relativistic effects arise from readout changes rather than spacetime deformation;
6. calculus and continuous fields arise from statistical limits of refresh differencing and have explicit failure boundaries.

From this chapter onward, invariance of c, relativity, and energy formulae are no longer empirical inputs, but direct readouts of the SU refresh structure.

Chapter 3: The Ontology of Charge: Phase Occupation, Not Material Properties

Goal of This Chapter

The goal of this chapter is to rewrite "charge" from an empirical term into an ontological variable within the NO5 framework, and to accomplish three key conclusions:

1. What is charge: it is no longer a "particle property," but a phase occupation within the SU refresh protocol.
2. Where do positive and negative charges come from: not from two separate hardware or space units, but from leading/lagging occupations within the same right-handed refresh protocol.
3. Why charge is quantized: it is not an experimental axiom, but an inevitable result of the minimal phase step in the space unit.

This chapter establishes the ontological concept of "electricity" and basic accounting rules, without introducing the formal derivation of "magnetic fields/Lorentz force/Maxwell's equations," though it provides the minimum interface definitions needed for subsequent chapters.

3.1 Why Charge Definition Must Be Rewritten

In traditional physics, charge is considered an inherent property of fundamental particles: it has magnitude, polarity, and is used as a source term in electromagnetic equations. While this approach works in engineering, it leaves three ontological gaps:

- Why does charge exist, and why is there symmetry between positive and negative charges?
- Why is charge quantized? This is only accepted as an experimental fact.
- Why is there no common underlying object between charge, "spatial structure," and "time readings"?

The NO5 strategy is:

Without introducing new entities (particles/fields/dimensions), charge is reduced to a phase variable in the refresh structure.

3.2 Phase Occupation: The Unique Takeaway of the NO5 Charge Concept

In Chapters 1 and 2, we have already established:

The space unit (SU) has a total refresh budget Ω_0, decomposed into two orthogonal phase directions:

- Stepping Phase ϕ_s: Migration (execution), cannot independently complete return.
- Settling Phase ϕ_r: Return (locking), is the only phase direction that can close.

This leads to the critical conclusion:

For charge to become an ontological variable, it must reside in the "phase structure," not in the "refresh speed."

The reason is quite straightforward:

Refresh speed belongs to the budget distribution (β_s, β_r), describing speed and energy readings; the polarity and quantization of charge cannot be directly generated by budget ratios but must arise from phase occupation.

Thus, the core object of this chapter is phase occupation.

3.3 Minimal Phase Step and the Definition of Charge

3.3.1 Minimal Phase Step $\Delta\phi$

In the discrete refresh protocol of SU, phase is not a continuous, arbitrarily adjustable "real number," but has a minimal resolvable phase step $\Delta\phi$. This is not an additional assumption, but the "minimum resolution" inherently brought by discrete refresh.

The minimal step $\Delta\phi$ represents the smallest non-zero deviation of phase that can occur while maintaining protocol consistency.

3.3.2 Ontological Definition of Charge

Definition 3.1 (Charge)

$$\boxed{q \equiv \pm\Delta\phi}$$

Here $\Delta\phi$ is not a continuously adjustable value, but the smallest distinguishable phase occupation label in the refresh protocol.

- Positive charge: Phase occupation $+\Delta\phi$ (leading).
- Negative charge: Phase occupation $-\Delta\phi$ (lagging).

It must be emphasized:

- Charge is not "how much phase," but "which phase position is occupied";
- Charge is not "how fast refresh happens," but "the minimal deviation relative to the background refresh sequence."

3.4 Why Positive and Negative Charges Don't Require Left-Hand Space Units

This is a crucial point that must be locked in this chapter.

The NO5 right-handed axiom stipulates:

All SU refresh sequences have a fixed right-handed orientation. Here, "right-handed" constrains the topological direction of the refresh sequence, not the "sign of phase values." Therefore:

- $+\Delta\phi$ and $-\Delta\phi$ are both realized within the same right-handed refresh sequence;
- $-\Delta\phi$ is not "left-handed refresh," but a "phase retreat branch" within the right-handed sequence.

Final conclusion: Negative charge does not come from left-handed space units or reverse refresh

protocols, but from a retreat in phase occupation within the same right-handed refresh sequence (-$\Delta\phi$). The right-handed constraint governs the refresh order but does not constrain the sign of phase occupation. This statement definitively resolves all doubts about the "existence of left-handed hardware."

3.5 Phase Codings: 0 / 1 / 9

To simplify the discussion and unify subsequent chapters (electron double-helix, nucleon layers, annihilation, etc.), phase codings are introduced:

- 0 bit: Standard background phase (no bias)
- 1 bit: Minimal leading occupation (+$\Delta\phi$)
- 9 bit: Minimal lagging occupation (-$\Delta\phi$)

This is a notation, not a new physical structure. Its key property is: $1 + 9 \equiv 0 \quad (mod 10)$

Physically, leading and lagging occupations in the same refresh sequence can fully self-cancel, restoring the background phase.

3.6 Why Charge Is Quantized: Not an Axiom, But Inevitable from Indivisible Phase Occupation

Traditional physics treats "charge quantization" as an experimental fact. NO5 provides a more ontological explanation:

- Phase occupation can only take discrete positions like 0, +$\Delta\phi$, and -$\Delta\phi$;
- There is no physical realization of "occupying 0.3$\Delta\phi$" because refresh and settling cannot recognize smaller units;
- Therefore, charge can only take discrete units of $\pm e$.

Final conclusion: Charge quantization is not an empirical input, but an inevitable result of the indivisibility of the minimal phase step.

3.7 Like Charges Repel, Unlike Charges Attract: Geometric Consequences of Self-Cancellation and Crowding

In the coding language:

- Opposite signs (1 and 9): $1 + 9 \equiv 0$
  This means the phase bias can fully self-cancel → the system tends toward this "low-residual" state → manifested as attraction.
- Same signs (1 and 1 or 9 and 9): $1 + 1 \not\equiv 0, \quad 9 + 9 \not\equiv 0$
  This means the bias cannot self-cancel, creating local phase crowding → the system tends toward separation to reduce crowding → manifested as repulsion.

The key point here: attraction/repulsion is not "forces pushing and pulling," but the phase system seeking a more easily returnable, lower-residual organization.

3.8 Neutrality and Annihilation: Ontological Expression of Charge Conservation

3.8.1 What Is Neutrality?

Neutrality is not "no phase," but the self-cancellation of leading and lagging occupations inside the enclosure structure, where the net occupation is 0.

In the coding scheme: $\sum q = 0 \quad \Leftrightarrow$ 1 and 9 appear in pairs

3.8.2 What Is Annihilation?

Positive-negative annihilation does not require "anti-matter mysticism." Ontologically, it is simply: occupation 1 and occupation 9 completing self-cancellation in the same enclosure/encounter structure, restoring the background 0.
After self-cancellation, phase still needs to be processed: if it cannot be settled within the local enclosure state, it transitions into a propagation state (electromagnetic waves/photon), which will be discussed in subsequent chapters.

3.9 Current, Voltage, Electric Field: Minimum Interface Definitions of This Chapter

To connect with Chapter 4, this chapter only provides the ontological interfaces for these quantities, without formal derivation.

3.9.1 Electric Current (current)

Definition 3.2 (Electric Current)
Current is not "charge particles running in a wire," but the continuous migration rate of phase occupation in the SU network.
Symbolically: $\mathbf{J} \equiv \rho_q \mathbf{u}$
where ρ_q is the occupation density, and $\mathbf{u}$ is the effective direction of occupation migration (depending on the specific material's phase-channel structure).

3.9.2 Electric Voltage

Definition 3.3 (Electric Voltage)
Voltage is the difference in "phase return conditions" between two locations, i.e., the difference in the cost required for phase occupation at the settling layer.
Voltage is not "energy difference," but the potential difference in how easily return can be achieved.

3.9.3 Electric Field

Definition 3.4 (Electric Field)
The electric field is the gradient reading of phase occupation in space, i.e., the rate of change of the "return potential" in space.

In the readout layer, it can be written using the phase potential ϕ: $\boxed{\mathbf{E} \equiv -\nabla\phi}$

However, it must be understood that in NO5, ϕ does not represent the "electric potential" of a continuous medium, but the encoding of phase occupation return conditions.

3.10 Why Macroscopic Objects Have No "Suspended Phase" but Can Still Carry Charge (Clarification)

Our latest discussion has established: macroscopic enclosure states do not have propagating "suspended phases" (internal immediate self-cancellation).
This does not deny macroscopic electrical properties because:

- Charge is occupation (static structure), not suspended propagation;
- The difference between macroscopic conductors and insulators is primarily whether occupation can migrate quickly, forming continuous migration in channels (current);
- Only when occupation migration or changes are too fast for internal closure to occur, will propagation states (radiation) emerge locally, which will be elaborated in subsequent electromagnetic wave chapters.

The macroscopic "no suspended phase" does not mean "no charge"; it simply means that the

propagation balance of occupation will be rapidly self-canceled within the enclosure, while electrical properties stem from the occupation itself, and electromagnetic effects arise from occupation migration and its closure-management structure.

3.11 Chapter Summary

This chapter completes the ontological reconstruction of the NO5 charge concept:

1. Charge is not a material property nor refresh speed, but phase occupation: $q = \pm\Delta\phi$.
2. Positive and negative charges correspond to leading/lagging occupation, without introducing left-handed space units; the right-handed constraint governs the refresh order, not the sign of phase occupation.
3. Charge quantization arises from the indivisibility of the minimal phase step.
4. Attraction/repulsion is a geometric consequence of self-cancellation and crowding of occupation, not "force" itself.
5. Current is occupation migration, voltage is return potential difference, and the electric field is the gradient of phase occupation.

Thus, "electricity" in NO5 is not an independent interaction, but the geometric manifestation of phase occupation and return conditions within the space unit's refresh protocol.

Chapter 4 Magnetic Field, Lorentz Force, and Maxwell's Equations: A Transparent Unification via the Phase-Closure Principle

Goal of This Chapter

This chapter proves that the magnetic field is not an independent entity, the Lorentz force is not a fundamental interaction, and Maxwell's equations are not a set of empirically juxtaposed postulates. All three arise from a single ontological principle: when the stepping phase continues to migrate and cannot self-close, the settling phase is forced to be used externally; the geometric organization of this external closure, the resulting trajectory correction, and the associated continuity constraints manifest respectively as the magnetic field, the Lorentz force, and Maxwell's equations.

4.1 Ontological Division of Phase Responsibilities (The Only Premise)

Recalling the previous chapter, the SU refresh budget decomposes into two functionally orthogonal phases:

$$\Omega_0^2 = \Omega_s^2 + \Omega_r^2, \quad \beta_s = \frac{\Omega_s}{\Omega_0}, \beta_r = \frac{\Omega_r}{\Omega_0}$$

- Stepping phase ϕ_s: migration/advance, cannot close;
- Settling phase ϕ_r: return/locking, the only phase that can close.

Closure can be completed only by ϕ_r. This is the sole ontological axiom of the chapter.

4.2 Ontological Definition of the Magnetic Field: External Closure of the Settling Phase

4.2.1 Trigger Condition (Not Merely "Having a Current")

The necessary and sufficient condition for a magnetic field to appear is: the stepping phase migrates persistently ($\phi_s \neq 0$), while the internal enclosure cannot immediately self-cancel the suspended residue. Consequently, the settling phase ϕ_r is forced to take over closure in the

external SU network.

4.2.2 Geometric Uniqueness (XYZ Coordinate Locking)

Adopt a three-dimensional Cartesian frame:

- Y-axis: migration axis of the stepping phase;
- X-axis: the radial return direction available in the stationary case;
- Z-axis: orthogonal to the X–Y plane.

When migration occurs along Y,

- radial closure along X is cut off within the same refresh tick;
- closure cannot be aligned with migration (otherwise it remains suspended);
- the only available closure path in 3D is a circumferential return around the Y-axis.

Definition 4.1 (Magnetic field).

$\mathbf{B}$ is the directional closed flow formed by the settling phase in the external SU network.

Its direction satisfies $\boxed{\mathbf{B} \parallel \hat{\mathbf{Y}} \times \hat{\mathbf{X}}}$. The right-handed orientation is uniquely fixed by the refresh order, not by empirical convention.

4.3 Lorentz Force: An Orthogonal Geometric Correction for Entering the Closure Channel

4.3.1 Ontological Cause

In a region with an externalized magnetic field (circumferential closure),

- an enclosure state carrying phase occupation travels straight along Y;
- a straight trajectory cannot enter the circumferential closure channel;
- without correction, closure fails (illegality).

4.3.2 The Unique Solution

To restore closure, the system must generate a geometric correction orthogonal to the migration axis, allowing the trajectory to connect to the circumferential closure.

Definition 4.2 (Lorentz force).

The Lorentz force is the orthogonal geometric correction imposed by the settling phase on the stepping-phase migration trajectory in order to restore closure.

Three properties follow directly (no extra assumptions):

- $\mathbf{F}_L \perp \mathbf{v}$ (orthogonality);
- it does no work (does not change β_s);
- it reverses sign for opposite occupations (different phase-occupation signs).

At the readout level, this is written as $\boxed{\mathbf{F}_L = q\,\mathbf{v} \times \mathbf{B}}$

The cross product is not a computational trick; it is the unique directional operator required to enter the closure channel.

4.4 Unified Ontological Origin of Maxwell's Equations

4.4.1 Readout-Level Definitions (Non-ontological)

$\mathbf{E} \equiv -\nabla\phi$ (radial gradient readout of phase occupation),

$\mathbf{B} \equiv$ orientation readout of the external closure of the settling phase.

4.4.2 Auditable "Hard-Core" Derivation on a Discrete SU Lattice

Discretize space into an SU lattice with spacing ℓ_0 and time step $\Delta t = \frac{\ell_0}{c}$ (the hardware

definition of "light-speed relay").
Electric field (edge readout): the discrete gradient of phase;
Magnetic field (face readout): the oriented flux through a minimal closure face.
The non-accumulation of closure residuals (the sole legality condition) yields the discrete Faraday update:

$$\boxed{(\nabla_h \times \mathbf{E})^n} = -\frac{\mathbf{B}^{n+1}-\mathbf{B}^n}{\Delta t} \tag{Fh}$$

The minus sign is jointly fixed by the right-hand edge-ordering and the zero-residual condition.

The discrete identity $\nabla_h \cdot (\nabla_h \times \cdot) = 0$ implies $\boxed{\nabla_h \cdot \mathbf{B}^n = 0}$ (no magnetic monopoles follow from difference algebra plus initial closure).
By symmetry of relay continuity (propagation state), one obtains

$$\boxed{(\nabla_h \times \mathbf{B})^n} = \frac{1}{c^2}\frac{\mathbf{E}^{n+1}-\mathbf{E}^n}{\Delta t} \tag{Ah}$$

Combining these yields the discrete wave equation: $\boxed{\frac{\mathbf{E}^{n+1}-2\mathbf{E}^n+\mathbf{E}^{n-1}}{\Delta t^2}} = c^2\,\Delta_h \mathbf{E}^n$

The propagation speed is uniquely fixed by the hardware step size to be c.

4.4.3 Continuous Limit (Standard Form)

Taking $\ell_0, \Delta t \to 0$ gives four equations as different projections of the same closure principle:

$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$ (occupation must be settled)

$\nabla \cdot \mathbf{B} = 0$ (closure cannot break)
$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ (radial change $\rightarrow$ circumferential reorganization)
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \partial_t \mathbf{E}$ (migration/variation must be carried into closure)

4.5 Electromagnetic Waves: Propagation Solutions of Settling-Loop Failure (Ontology)

When phase variation or migration makes local settling closure unsustainable,

- the settling phase withdraws;
- the stepping phase takes over;
- the phase pattern is relayed through the SU network at light speed.

Electromagnetic waves are not mutual excitation between $\mathbf{E}$ and $\mathbf{B}$, but the propagation-state ontological equation $\boxed{\partial_t \phi + c\,\hat{\mathbf{n}} \cdot \nabla \phi = 0}$

$\mathbf{E}$ and $\mathbf{B}$ are only two orthogonal readout constraints of this propagation state.

4.6 One "Total Constraint" Governs the Whole Chapter

Total phase-closure constraint (NO5):
occupation cannot remain suspended($\nabla \cdot \mathbf{E} - \rho/\varepsilon_0 = 0$);
closure cannot break ($\nabla \cdot \mathbf{B} = 0$);
variation must be entrusted ($\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$);
migration/variation must be carried ($\nabla \times \mathbf{B} - \mu_0 \mathbf{J} - \mu_0 \varepsilon_0 \partial_t \mathbf{E} = 0$).
These four are not parallel postulates but projections of a single legality condition.
Ontological status and geometric appearance of the magnetic flux tube

In NO5, only the settling phase can close; the stepping phase migrates and cannot close by itself. Any stable or repeatable structure must ultimately be returned by the settling phase. A magnetic flux tube is not a new phase degree of freedom and not an independently closable object.

When there is no directed stepping migration, settling can close locally in an approximately isotropic manner, without forming a directional structure. Once sustained, directed stepping migration exists (e.g., currents or internal advancement within an enclosure state), the settling phase faces a hard constraint: it cannot close along the stepping direction, otherwise closure would be immediately dragged away and fail.

The only feasible choice is therefore to form a return path in the plane orthogonal to the stepping direction. This return path must be continuous, repeatable, and self-canceling tick by tick. The minimal geometric realization is an annular return channel around the stepping axis. The magnetic flux tube is the stable statistical manifestation of this return channel.

Hence, the magnetic flux tube should be understood as the geometric return mode that the settling phase is forced to adopt in order to maintain closure consistency when directed stepping migration is present. It is not the "agent of closure," but an accompanying structure of how closure occurs. Its stable, trackable appearance arises from the statistical accumulation of return density over many refresh cycles, not from an independent ontological status.

Moreover, because settling closure must self-cancel tick by tick, any one-way circulation would leave a net residue. Therefore, the return path must statistically split into oppositely oriented circulation components. In this sense, a flux tube appears as a double-helical pattern in spacetime: not two independent structures, but the left–right counter-winding projection of the same settling return channel under continuous refresh. The "helical" appearance comes from the geometric superposition of refresh advance and return paths, without introducing any new physical object.

Accordingly, a flux tube carries no propagation speed, transmits no independent information, and does not participate in dynamics beyond settlement. It is strictly distinct from electromagnetic waves (propagation states after closure failure): the former is a geometric manifestation of closure mode, whereas the latter is the expulsion result of closure failure. This distinction explains the endpoint-free nature of flux tubes and rules out magnetic monopoles at the ontological level.

In summary, the magnetic flux tube is not a third kind of physical entity, but an inevitable accompanying geometric structure generated when the settling phase is forced to operate externally under sustained stepping migration. This secures the ontological closure and self-consistency of the unified derivation of magnetic fields, Lorentz force, and Maxwell's equations in this chapter.

4.7 Conclusions

1. The magnetic field is the directional closure formed by the forced external use of the settling phase;
2. The Lorentz force is the orthogonal geometric correction required to enter the closure channel;
3. Maxwell's equations are the four projections of phase-closure legality;
4. Electromagnetic waves are propagation solutions after settling-loop failure;
5. right-handedness, the minus sign, the absence of magnetic monopoles, and propagation speed c are all rigidly derived from consistency of the discrete SU lattice.

4.8 Generator: The Minimal Engineering Device that Forces External Use of the Settling Phase

4.8.1 Ontological definition of generation (NO5)

Definition 4.8 (Electric power generation).

Generation is not "a wire cutting magnetic field lines," but mechanically forcing the breakdown of an existing settling-closure structure, thereby forcing the settling phase to be used externally and converting it into a phase-occupation flow (current) that can migrate along a conductor channel. This is the mother definition of all engineering applications in Chapter 4.

4.8.2 Ontological process of a rotating-coil generator (step-by-step)

Consider the simplest generator model: a rigid coil rotating about an axis in a stable externalized magnetic field. In NO5 language, what occurs is not "a change of magnetic flux," but:

1. the internal settling closure of the coil is originally statically achievable;
2. mechanical rotation changes the coil orientation in time;
3. the original circumferential settling closure ceases to be self-consistent within a refresh cycle;
4. the settling phase is forced to be used externally and searches for a new closure along the conductor path;
5. this externalized settling phase manifests in the conductor channel as directed migration of phase occupation—current.

4.8.3 Ontological rewriting of Faraday's law

Traditional form: $\mathcal{E} = -\frac{d\Phi_B}{dt}$

NO5 ontological form: $\mathcal{E} \equiv \oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt}$ [available settling-phase closure] (4.18)

The minus sign does not mean a "direction convention," but that the system always rearranges phase toward restoring closure.

4.8.4 Minimal conclusion for generators

A generator = mechanically forcing stepping motion to break settling closure → forced external use of the settling phase → current appears. No need for:

- an "energy-conversion" assumption;
- field-line ontology;
- a "push" by fields.

4.9 Transformer: Geometric Reconstruction of the Settling-Phase Closure Path

4.9.1 Ontological statement

Definition 4.9 (Voltage transformation).

Transformation is not "a magnetic field transferring energy from one side to the other," but a redistribution of the settling phase into different geometric channels within a shared closure core (the iron core).

4.9.2 What happens in the primary coil

When AC is applied to the primary:

- phase occupation migrates periodically;
- the settling phase is periodically externalized;
- a highly coherent, low-dissipation closure channel forms in the core.

The role of the core is not to "enhance the magnetic field," but to provide a stable closure geometry with low settlement impedance.

4.9.3 Why the secondary must respond

Due to the sharing of the same settlement closed core between the primary and secondary levels:

- reorganization of settlement phase in primary level
- will inevitably change the phase availability distribution of the entire closed structure

To maintain closure consistency, the system can only externalize the settling phase in the secondary, producing current.

4.9.4 Ontological origin of the turns ratio

Traditional result: $\frac{V_2}{V_1} = \frac{N_2}{N_1}$

NO5 interpretation:

- each turn does not "increase flux," but increases one sampling count of available settling closure;
- the turns ratio equals the geometric allocation ratio of the settling phase.

Thus:

$$\frac{N_2}{N_1} \text{ Settlement Closure Sampling Count }_1,\ \frac{V_2}{V_1} \text{ Settlement Closure Sampling Count }_2 \qquad (4.19)$$

4.9.5 Minimal conclusion for transformers

A transformer = redistributing the geometric weights of the settling phase within a shared closure core.

4.10 R, L, C: Three "Impedance Forms" of the Settling Phase

(This section is the engineering "crown" of Chapter 4.)

4.10.1 Resistance R: Dissipative impedance of the settling phase

Definition 4.10 (Resistance).

Resistance is not "electron collisions," but the dissipative impedance produced when the settling phase is forced to traverse many non-uniform closure paths in a material.

Readout: V = RI

Ontological meaning: externalized settling phase → repeated failed micro-closures → conversion into local thermal settlement (temperature rise).

4.10.2 Inductance L: Inertia of the settling-closure structure

Definition 4.11 (Inductance).

Inductance is the inertia of the settling-phase closure structure against change.

When current changes:

- the closure structure must be globally rearranged;
- rearrangement cannot be instantaneous;
- thus the system resists change.

Readout: $V_L = L\frac{dI}{dt}$

Ontological meaning: this is geometric inertia of closure, not "energy stored in the magnetic field."

4.10.3 Capacitance C: Storage capacity of the settling phase

Definition 4.12 (Capacitance).
Capacitance is the ability of a system to temporarily store the settling phase before closure is completed.
Between capacitor plates:

- phase occupation is separated;
- closure is delayed;
- the system is in an "unsettled tension state."

Readout: $Q = CV, \quad I_C = C\frac{dV}{dt}$

4.11 RLC: Full Dynamics of the Settling Phase

4.11.1 Unified RLC equation

Traditional form:

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C}V(t)$$

NO5 ontological reading: settlement inertia + settlement dissipation + settlement tension + external settlement driving. (4.20)

4.11.2 Ontological meaning of resonance

RLC resonance is not "energy shuttling between L and C," but the rhythm of externalization–recovery of the settling phase matching the system's geometric closure period.

Resonance frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$

interpreted as:

- settlement storage time (C);
- closure-rearrangement inertia (L);
- achieving a minimum-residue cycle.

4.11.3 Phase lag and phase lead

AC phase phenomena:

- inductive lag;
- capacitive lead;

are unified in NO5 as differences in the temporal rearrangement of the settling phase relative to the stepping phase.

4.12 Supplementary Summary of Chapter 4 (Engineering Unification)

1. Generator: mechanical stepping → breaking settling closure → current emerges.
2. Transformer: shared settling-closure core → redistribution of geometric weights.
3. R, L, C:

- R: settlement dissipation;
- L: settlement inertia;
- C: settlement storage.

4. RLC: a complete dynamical system of the settling phase.

All electromagnetic engineering devices are geometric realizations of how to externalize, store, rearrange, and dissipate the settling phase. Once the magnetic field is understood as externalized closure of the settling phase, generation, transformation, and RLC are no longer mere applications

of electromagnetism; they become natural engineering extensions of NO5 ontology. In NO5, electromagnetism is reduced to the inevitable manifestation of “phase-duty allocation and closure legality,” rather than an independent interaction system.

Chapter 5: The Minimum Stable Enclosure of the Electron: The Necessity of the Right-Hand Double-Helix and Phase Cancellation

Goal of This Chapter

This chapter addresses an unavoidable question within the NO5 framework:
What is the minimum structure that can exist stably for a long time, under a refresh protocol that only allows "stepping-phase migration and settling-phase closure"?
The conclusion will prove that:
The electron is the first stable enclosure state "allowed" in NO5, and its structure is not an assumption, but the minimal solution uniquely chosen by the necessity of phase closure.

5.1 Ontological Criterion for Stable Enclosure (Establishing the Rules)

Reviewing the ontological division from Chapter 1:

- Stepping phase ϕ_s: responsible for migration, cannot close
- Settling phase ϕ_r: responsible for return, the only phase that can close

Definition 5.1 (Stable Enclosure State)
An enclosure state is considered stable if and only if:

1. The phase residual $\delta\phi_s$ generated by the stepping phase during any refresh cycle can be fully reclaimed by the settling phase;
2. The reclaiming process does not depend on the external SU network (otherwise it is a propagation state);
3. The return path is repeatable in time (closure does not drift).

This is a criterion for existence, not a dynamical approximation.

5.2 The Necessity of a Single-Loop Enclosure (Proof by Elimination)

Imagine the simplest enclosure:
A single closed loop, carrying a phase occupation $q = \pm\Delta\phi$.
In the right-handed refresh sequence:

- The stepping phase migrates circumferentially;
- Each refresh tick produces phase migration $\delta\phi_s$;
- The settling phase attempts to close along the same path.

However, due to:

- The orthogonality of settling and stepping responsibilities
- The single-loop structure cannot complete "migration and return" in the same cycle

Phase drift $\sum_{cycle} \delta\phi_s \neq 0$ inevitably arises, leading to:

- The enclosure center wanders in the SU network
- Stability is broken

Theorem 5.1 (Instability of Single-Loop Enclosures)
Under the NO5 refresh protocol, any enclosure state containing only a single-phase loop cannot satisfy the stable enclosure criterion.

5.3 The Emergence of the Double-Loop Structure: The Only Remedy

To eliminate the drift, the system must introduce within the same enclosure:

- A $+\Delta\phi$ migration branch
- A $-\Delta\phi$ return branch

And require that both:

- Belong to the same right-handed refresh sequence
- Self-cancel phase residuals within each refresh cycle

This directly leads to the minimum structure:

A pair of right-handed closed loops, each carrying $+\Delta\phi$ and $-\Delta\phi$.

5.4 Phase Codings and Internal Cancellation Conditions (Mathematical Closure)

Using the phase coding from Chapter 3:

- $1 \equiv +\Delta\phi$
- $9 \equiv -\Delta\phi$

The double-loop structure satisfies: $1 + 9 \equiv 0 \pmod{10}$. This is an internal cancellation condition that holds tick by tick, not an average over a cycle.

Theorem 5.2 (Minimal Internal Cancellation)

The double-loop (1,9) is the minimal structure in NO5 that achieves phase residual cancellation tick by tick. Any fewer structures cannot eliminate drift, and any more are redundant.

5.5 Right-Handedness and the Strict Distinction from "Reverse Loops"

A critical misunderstanding must be clarified here.

- Right-handedness constrains the topological direction of the refresh sequence.
- The sign of the phase describes occupation values.

Therefore:

- $+\Delta\phi$ and $-\Delta\phi$
- Are not "left-handed/right-handed hardware"
- But leading/lagging occupations in the same right-handed sequence

The "reverse" in the double-loop is not reverse refresh but the opposite symbol of phase occupation within the right-handed refresh sequence. This step guarantees that the electron model is consistent with NO5's basic axioms.

5.6 Spin (1/2): The Direct Result of the Closure Topology

In the double-loop structure:

- A single 2π phase return → makes only one loop return to its initial state;
- Both loops returning to the initial state → requires a 4π return.

Mathematically: $\psi(\theta + 2\pi) = -\psi(\theta), \psi(\theta + 4\pi) = \psi(\theta)$

Theorem 5.3 (Origin of Spin)

The electron's spin (1/2) originates from the 4π closure topology of the right-handed double-loop, not from internal true rotation.

5.7 Intrinsic Magnetic Flux Tube: Internal Closure of the Settling Phase

In the double-loop structure:

- The settling phase forms a stable closure along the loop

- No external SU network is required to take over
- This closure has directionality

This directional internal closure: $\mu \propto \oint \phi_r \, d\ell$ is the ontological source of the electron's magnetic moment.

Important Distinction:

- The intrinsic magnetic flux tube maintains existence and does not manifest when stationary;
- The external magnetic field (Chapter 4) is the settling phase externalized, requiring migration to trigger.

The double-helix is not an additional structure, but the geometric projection of the same closure return channel in time expansion.

5.8 Why the Electron Does Not Radiate When Stationary

Radiation in NO5 is defined as:

When an enclosure state loses its settling-closure, it is forced into the propagation state.

For the electron double-loop:

- The residuals of the stepping phase self-cancel tick by tick;
- The settling phase completes closure entirely internally;
- No external use of settling is needed.

Thus: $\delta\phi_s^{cycle} = 0 \quad \Rightarrow$ no propagation state arises

Theorem 5.4 (Non-Radiative Property)

The electron does not radiate in a stable enclosure state, which is a direct consequence of phase self-cancellation, not a dynamical accident.

5.9 Hierarchical Positioning and Irreplaceability of the Electron

Based on the results above, the electron has the following irreplaceable features:

1. It is the minimal stable enclosure (no smaller structure can remain stable);
2. It uniquely requires a double-loop (a single loop is unstable, and a triple loop is redundant);
3. Its spin 1/2 has the only topological source;
4. Its magnetic moment and charge are naturally unified;
5. It is the foundational module for all higher layers (nucleons, atoms).

5.10 Conclusion

1. Stable enclosure requires phase residuals to self-cancel tick by tick;
2. A single-loop enclosure inevitably drifts and cannot remain stable;
3. The right-handed double-loop (1,9) is the minimal stable solution;
4. Spin 1/2 originates from the 4π closure topology;
5. The electron's magnetic moment arises from the internal closure of the settling phase;
6. The electron does not radiate because the closure does not fail.

Thus, the electron is not a fundamental particle, but the minimal enclosure structure "allowed" in NO5.

5.11 Clarification: Equivalence of "Double-Helix" and "Double-Loop" Descriptions and Theoretical Upgrade

In earlier works in this series (prior Natural Origins papers), the electron's minimal stable

enclosure state was described as a double-helix structure. In this paper, the electron is characterized as the minimal stable enclosure formed by a pair of right-handed double-loops. It's important to emphasize that this change is not a model rejection or self-correction but an equivalent reconstruction and expression upgrade following the deepening of the theory.

(1) Why Switch from "Double-Helix" to "Double-Loop"

The core purpose of the early "double-helix" description was to emphasize:

1. Right-handedness: the electron's enclosure state follows a unified right-handed refresh sequence;
2. Phase counter-winding: two phase branches ($+\Delta\phi$ and $-\Delta\phi$) participate in enclosure to achieve self-cancellation tick by tick.

In the original context, the term "helix" was used to intuitively describe the continuous circumvolution of the phase as the refresh process advances. However, with the rigorous formalization of the orthogonal phase of spatial elements and the refresh-closure structure in this paper, further analysis reveals that:

- the key to electronic stability does not rely on the "axially stretched pitch morphology",
- but on the closed topology of phase occupancy and the intra-beat cancellation condition.

At this more ontological level, "double loop" serves as a more precise and minimal topological representation: it directly corresponds to closed paths and intra-phase cancellation, and avoids unnecessary commitments to geometric morphologies (e.g., pitch, stretching direction).

Therefore, the "double helix" and the "right-handed double loop" are physically equivalent; the latter is merely a topological compression and ontological abstraction of the former.

(2) "Double-Loop" Does Not Deny Dynamic Phase Progression

It must be clarified that the "double-loop" description does not imply that the phase is stationary or not progressing.

In the NO5 framework:

- Phase progression is carried out by the stepping phase;
- Closure and existence are completed by the settling phase;
- The double-loop merely describes the minimal topological unit of the closure return path, not negating phase flow in the refresh process.

In other words, the "helix" is not negated, but absorbed into the refresh process itself. "Loop" is only used to indicate the topological fact of closure and self-cancellation.

(3) The Same Applicability for Proton and Neutron Models in Later Chapters

The same clarification applies to the subsequent descriptions of protons and neutrons in this paper. In the preceding work, the nucleon model was presented in the more concrete terminology of a "multi-helix encapsulation"; in this paper, this description is systematically elevated to the second stable encapsulation level constituted by three pairs of orthogonal right-handed double loops.

This modification is likewise not a repudiation of the original model, but rather:

- a minimized, orthogonalized and theorematic formulation of the original structure,
- following the introduction of the three-dimensional stability criterion and orthogonal phase decomposition.

The nucleon model proposed herein retains all the physical connotations of the preceding work (right-handedness, phase counter-rotation, and encapsulation hierarchy), and provides a more rigorous mathematical and topological definition thereof.

(4) Theoretical Evolution Principle

In summary, the changes in the description of the electron, proton, and neutron in this paper should be understood as:
A stepwise abstraction, compression, and formalization of the same physical structure under the unified space-unit refresh framework, not a change in theoretical stance.
This evolution follows the principle that when a physical structure can be expressed with fewer geometric assumptions and clearer topological conditions, the latter should be prioritized.
The change in how the electron and nucleon enclosures are described reflects the natural upgrade in theory after the introduction of space-unit orthogonal phases and refresh-closure criteria, rather than a denial of earlier models.

Chapter 6 Protons and Neutrons: The Second Stable Enclosure Level of Three Mutually Orthogonal Double Loops

Goal of This Chapter

In Chapter 5 we proved that the right-handed double-loop (1–9) is the minimal stable enclosure state permitted to persist in NO5 (the electron). This chapter addresses a deeper question: under the same refresh protocol, does there exist an enclosure state that is higher-order than the electron yet still stable? If so, how are its structure, mass scale, and electric character forced by phase closure?
The conclusion will show that protons and neutrons are not independent "new particle assumptions," but the second stable enclosure level in NO5, whose unique solution is three mutually orthogonal right-handed double loops.

6.1 Necessity of an Enclosure-Level Transition

The electron's double-loop enclosure satisfies the stability criterion, but it still has two fundamental limitations:

1. Geometric dimensional limitation
- the double loop closes only within a single plane;
- it is only "conditionally stable" against fully three-dimensional perturbations.

2. Settlement-capacity limitation
- the internal capacity of the settling phase is limited;
- it cannot serve as a stable anchor for multiple external enclosures (e.g., multiple electrons).

Therefore, if nature admits an "atomic-nucleus–level" stable center, its enclosure state must: accomplish tick-by-tick cancellation of stepping residuals simultaneously in three dimensions, and possess higher settlement-bearing capacity.
This is not an empirical demand, but a necessary functional upgrade of enclosure.

6.2 Geometric Criterion for Three-Dimensional Stable Enclosure

Recall the stability criterion (Chapter 5):
- stepping residuals must self-cancel tick by tick;
- the closure must not rely on the external SU network;
- the closure path must be temporally repeatable.

In three dimensions, if closure occurs only in a single plane, then any stepping perturbation perpendicular to that plane will leave an unrecoverable residual within one cycle.

Theorem 6.1 (Necessary condition for three-dimensional stability).
For an enclosure state to be unconditionally stable in three-dimensional space, it must possess closure capability simultaneously along three pairwise orthogonal directions.

6.3 Uniqueness of Three Orthogonal Double Loops (Elimination Argument)

In NO5, the minimal unit that can cancel stepping residuals internally is:

- one pair of right-handed double loops (1–9).

To satisfy Theorem 6.1 in three dimensions, the system must introduce:

$(1_x, 9_x)$, $(1_y, 9_y)$, $(1_z, 9_z)$

i.e., three pairs (six loops), each completing tick-by-tick cancellation in one of three orthogonal planes.

Excluding other possibilities

- fewer than three pairs: → at least one direction lacks closure → instability;
- more than three pairs: → no added stability, only increased internal tension → redundancy;
- non-orthogonal arrangement: → mutual interference of closure paths → failure of tick-by-tick cancellation.

Theorem 6.2 (Uniqueness).
Three pairwise orthogonal right-handed double loops constitute the unique minimal structure that realizes three-dimensional stable enclosure in NO5.

6.4 Ontological Definition of the Proton

Definition 6.1 (Proton).
A proton is a three-dimensional stable enclosure state composed of three orthogonal right-handed double-loop pairs. Its phase occupation self-cancels tick by tick along all directions, yet the structure retains an externally readable electric occupation interface.

This "external interface" means:

- all six internal loops participate in stability;
- the electric character is not fully neutralized internally;
- hence it manifests externally as charge $+e$.

6.5 Ontological Definition of the Neutron: A Phase-Compensation State

Relative to the proton structure, a key modification is introduced: the externally readable electric interface is also absorbed into the internal cancellation structure. Concretely:

- in addition to the three orthogonal double-loop pairs,
- an additional electric-compensation occupation structure is included,
- so that the external phase occupation vanishes completely.

Definition 6.2 (Neutron).
A neutron is a three-dimensional stable enclosure state composed of three orthogonal right-handed double-loop pairs, whose external phase occupation is fully compensated and thus appears electrically neutral.

Importantly:

- the neutron is not "proton + electron";
- it is a reorganization of internal structure within the same enclosure level.

6.6 Hierarchical Origin of the Mass Scale (Core Derivation)

In NO5, mass is not an input constant, but a readout of the settling-phase capacity that must be invested to guarantee tick-by-tick cancellation.

6.6.1 Mass and enclosure scale

Electron: 1 double-loop pair → 2 closure channels.

Proton/neutron: 3 double-loop pairs → 6 closure channels.

However, mass growth is not a linear multiple; it includes a hierarchical transition factor.

6.6.2 Second visibility threshold: α^2

As the first enclosure level, the electron's electric/magnetic visibility has already been "diluted" once. Entering the second enclosure level, visibility must pass through the same channel structure again: visibility strength $\propto \alpha^2$. This is a second filtering of structural visibility, not a coincidental appearance of an electromagnetic coupling constant.

6.6.3 Second closure threshold: π^2

One full closure of the electron double loop requires 2π. Three pairs of double loops constitute "closure of closure," so a stability threshold naturally introduces $(2\pi)^2 \sim \pi^2$ as an unavoidable cost of a higher geometric closure level.

6.6.4 Structural formula for the mass ratio

Combining the two ingredients yields: $\frac{m_p}{m_e} \approx \frac{\eta_\Omega}{\alpha^2 \pi^2}$

where: η_Ω is the geometric-consistency correction factor (order unity, not a free parameter) for three orthogonal double-loop pairs under the actual refresh protocol.

This is not a fitting formula, but a hierarchical structural relation.

6.7 Unified Origin of Magnetic Moment and Spin

Electron: single-plane double loop → spin 1/2.

Proton/neutron: three-plane double loops → externally still spin 1/2.

Reason: spin is not "the number of internal rotations," but a return property of the outermost closure topology. Although three double-loop pairs are present, only one effective topological return structure is externally exposed.

6.8 Ontological Explanation of Stability, Lifetime, and Nuclear Scale

- Electron: absolutely stable (minimal level).
- Proton: extremely stable (three-dimensional internal cancellation, stable external interface).
- Neutron:
  in the free state, it contains internal phase-compensation tension;
  when enclosed in a nucleus, it can be stabilized by the external settlement field.

Theorem 6.3 (Neutron lifetime).

The neutron lifetime is not an accidental result of the weak interaction, but the inevitable relaxation of a phase-compensation state after losing its external settlement anchor.

6.9 Direct Link to Black Holes and the Density-Consistency Theorem

Three orthogonal double-loop pairs imply:

- enclosure density has a geometric upper bound;
- adding more enclosure can only enlarge volume, not increase density.

This provides a microscopic basis for the density-consistency theorem in Chapter 13: a black hole is not "infinite compression," but a limit state where enclosure hierarchy reaches the dual extreme of stepping closure and settlement capacity.

6.9.1 Definition: η_Ω is not a parameter, but a minimum-residual minimizer

For the minimal proton enclosure family $\mathcal{G}$ (three orthogonal double-loop pairs + single channel + right-handed refresh + non-crossing), define the closure-residual functional:

$$\mathcal{F}(g) = \sum_{i\in x,y,z} \varepsilon_i\,(g)^2 + \lambda_\perp\,\mathcal{P}_\perp(g) + \lambda_\times\,\mathcal{P}_\times(g), \quad g \in \mathcal{G} \tag{6.9-1}$$

Here:

- ε_i: net residual of "1/9 cancellation" for the (i)-th double-loop pair (ideal 0);
- $\mathcal{P}_\perp$: penalty for orthogonality deviation (ideal 0);
- $\mathcal{P}_\times$: penalty for non-crossing / minimum separation (ideal 0);
- $\lambda_\perp, \lambda_\times$: weights that quantify hard constraints (large values correspond to the hard-constraint limit).

Define the unavoidable minimal closure-failure degree:

$$\delta_\Omega \equiv \min_{g\in\mathcal{G}} \mathcal{F}\,(g) \tag{6.9-2}$$

Define the geometric-consistency correction factor:

$$\boxed{\eta_\Omega \equiv \frac{1}{1+\delta_\Omega}} \tag{6.9-3}$$

Thus, once $\delta_\Omega > 0$ is established and a lower bound is estimated, the numerical range of η_Ω follows.

6.9.2 Minimal realizable shear angle (θ_0) for three orthogonal pairs

In an ideal mathematical model, three double-loop pairs could be perfectly orthogonal without interference. In NO5 physical enclosure, two additional hard constraints must also be satisfied:

1. single-channel constraint: all three pairs must share the same principal closure channel (the same "settlement return axis / flux-tube bundle"); otherwise multi-channel readout breaks single-valued stability;
2. non-crossing constraint: finite effective tube diameter implies geometric conflict at orthogonal intersections unless a minimal shear angle allows layered passage.

At minimal complexity, treat the three pairs near the shared axis as three orthogonal "closure bands." To avoid crossing while preserving symmetry, all three must adopt the same minimal shear angle θ_0. The symmetric minimal shear is given by equal partitioning:

$$\tan\theta_0 = \frac{1}{\sqrt{3}} \quad \Rightarrow \quad \boxed{\theta_0 = 30^\circ} \tag{6.9-4}$$

Meaning: for three orthogonal closures to share a single channel, a minimal three-direction shear partition near the axis is required; the symmetric minimal solution is a uniform 30° shear.

6.9.3 Minimal closure-failure degree δ_Ω from the shear angle

The shear does not break the right-handed refresh order nor the "1/9 cancellation" sign structure, but it implies:

- each pair's tick-by-tick closure path gains a minimal detour relative to the ideal path;
- detours cannot fully cancel at the shared-channel junction, leaving a net closure deficit.

For small shear, expand to lowest nontrivial order (keeping only geometrically necessary terms):

$$\delta_\Omega \simeq \underbrace{\frac{1}{8}\sin^2\theta_0}\ 1st-order\ detour + \underbrace{\frac{1}{32}\sin^4\theta_0}\ 2nd-order\ junction\ curvature + \underbrace{\epsilon_{tube}}_{finite\ tube\ diameter\ /\ buffer\ margin}$$

(6.9-5)

The first two terms are the minimal detour and curvature costs at the shared junction; $\epsilon_{\text{tube}} > 0$ is the minimal buffer required by finite tube diameter to enforce non-crossing and stable closure. Substituting $\theta_0 = 30°$ （$\sin^2\theta_0 = 1/4,\ \sin^4\theta_0 = 1/16$）：

$$\delta_\Omega \simeq \frac{1}{8}\cdot\frac{1}{4} + \frac{1}{32}\cdot\frac{1}{16} + \epsilon_{tube} 0.03125 + 0.001953 + \epsilon_{tube} \tag{6.9-6}$$

For the minimal realizable enclosure, take a conservative structural value:

$$\epsilon_{tube} \approx 0.0020 \pm 0.0010 \tag{6.9-7}$$

then

$$\delta_\Omega \approx 0.0352 \pm 0.001 \tag{6.9-8}$$

hence

$$\eta_\Omega = \frac{1}{1+\delta_\Omega} \approx \frac{1}{1.0352} \approx 0.966 \tag{6.9-9}$$

This is exactly the geometric-consistency correction scale required by the mass-ratio structure formula.

6.9.4 Closure with the proton–electron mass ratio

Substituting (6.9-9) into the structural relation

$$\frac{m_p}{m_e} \approx \frac{\eta_\Omega}{\alpha^2\pi^2} \tag{6.9-10}$$

yields the leading-order prediction for m_p/m_e. Here η_Ω is not "hand tuning," but is determined jointly by:

- three orthogonal double-loop pairs
- single-channel constraint
- non-crossing constraint
- right-handed refresh order

through the minimal shear angle and minimal closure-failure degree.

6.9.5 Remark: Falsifiability and auditability

This derivation of η_Ω has two clear falsification routes:

1. if future numerical minimization of geometric parameterizations under "three orthogonal + single channel + non-crossing" yields θ_0 significantly different from $30°$, this derivation must be updated;
2. if a geometry satisfying all hard constraints with $\delta_\Omega = 0$ can be constructed within the same framework, then η_Ω must return to 1, falsifying this subsection.

Thus η_Ω is a computable geometric quantity rather than an empirical patch; its value 0.966 arises from the structural necessity of minimal-shear closure.

6.9.6 Ontological equivalence to the Natural Origin IV correction (brief)

The geometric-consistency factor η_Ω obtained here is ontologically equivalent to the correction

coefficient introduced in Natural Origin IV via the "difference between proton light-speed closure and background light-speed flow." In Natural Origin IV, the coefficient arises because the proton, as a light-speed closed-loop enclosure, must remain stable within a background space that also refreshes at light speed; the enclosure loop cannot perfectly coincide with the background flow, hence an irreducible minimum efficiency mismatch appears, described dynamically as a "minimum mismatch between the light-speed loop and background light-speed."
NO5 elevates this result in hierarchy: by introducing orthogonal SU phases, three orthogonal double-loop pairs, and hard constraints of single channel and non-crossing, the "minimum mismatch" is rewritten not as a speed comparison but as a minimal closure residual δ_Ω, and η_Ω is defined accordingly. Hence the "light-speed difference" language and the "geometric closure residual" language describe the same physical fact at two equivalent levels: the former as dynamical intuition, the latter as a theoremized geometric–topological statement. The present work does not negate earlier conclusions; it upgrades them into a structural result independent of specific parameterization.

6.10 Chapter Conclusions

1. Three-dimensional stable enclosure requires tick-by-tick cancellation along three orthogonal directions;
2. three orthogonal right-handed double-loop pairs form the unique minimal solution;
3. the proton is a three-dimensional enclosure state that retains an external electric interface;
4. the neutron is a three-dimensional enclosure state that completes electric compensation;
5. the mass scale is set by the second visibility threshold α^2 and the second closure threshold π^2;
6. the black-hole density-consistency theorem gains microscopic support at this level.

Therefore, protons and neutrons are not "new fundamental particles," but necessary consequences of enclosure-level transitions in NO5.

Chapter 7 Atomic Energy Levels and Orbits: An Integer Spectrum from Pure Stepping-Phase Closure

Goal of This Chapter

The goal of this chapter is neither to "explain the Bohr model" nor to "patch quantum mechanics," but to answer a more fundamental question:
Under the NO5 refresh protocol, what kinds of stepping motion can persist over long times?
The answer will be given by a strict closure integral condition, from which the following results emerge naturally:

- the $1/n^2$ spectral law of atomic energy levels;
- quantization of orbital angular momentum;
- the common origin of atomic and planetary orbits.

No potential-energy hypothesis is introduced, no wavefunction is assumed a priori, and no classical-mechanics equilibrium is used.

7.1 The Closure Functional of the Stepping Phase

Recall from Chapter 6 that any stable enclosure state must achieve tick-by-tick self-cancellation.

For an "orbital state," this requirement is expressed geometrically as full-period closure of the stepping phase. Define the stepping-phase functional along a closed path C:

$$\Phi_s[C] \equiv \oint_C \frac{v(\mathbf{r})}{c} \frac{ds}{\ell(\mathbf{r})} \tag{7.1}$$

Here:

- ds: line element along the path;
- $\ell(\mathbf{r})$: local stepping-closure scale;
- $v(\mathbf{r}) = \beta_s(\mathbf{r})c$: stepping linear speed.

Ontological stability criterion:

$$\Phi_s[C] = 2\pi n, \quad n \in \mathbb{Z} \tag{7.2}$$

This is not a "quantum postulate." If the stepping phase fails to close, the settling phase must be used externally and the orbital state becomes unstable.

7.2 Minimal Closure Solution for a Circular Orbit

Consider first the most symmetric case: $r = const, \quad v = const$

Then (7.1) reduces to

$$\Phi_s \oint \frac{v}{c} \frac{ds}{\ell} \frac{v}{c} \frac{2\pi r}{\ell}$$

Imposing the closure condition (7.2) gives

$$\frac{v}{c} \frac{r}{\ell} = n \tag{7.3}$$

This is the NO5 minimal-orbit equation.

7.3 Atomic Scale: Determination of ℓ and the Fine-Structure Constant

In atomic problems, the stepping-closure scale ℓ is not a free parameter; it is fixed by the electron–nucleus enclosure structure. From the enclosure-hierarchy analysis in Chapter 6:

$$\ell = \frac{\hbar}{m_e c} \tag{7.4}$$

Substituting (7.4) into (7.3) and using $v = \beta_s c$, we obtain

$$m_e v r = n\hbar \tag{7.5}$$

This is orbital angular-momentum quantization, but in NO5 it is not assumed—it is the direct algebraic result of the closure condition.

7.4 The Settling-Phase Anchor and the Radial Constraint

An orbit requires not only angular closure but also radial "settleability." In NO5, the Sun or an atomic nucleus acts as a settling-phase anchor, described by the settling-fraction field $\beta_r(r)$. The radial stability condition (in the macroscopic form used in Chapter 9) is

$$\frac{v^2}{r} c^2 \frac{d}{dr} \ln \beta_r(r) \tag{7.6}$$

At atomic scales, the weakest-field expansion is $\beta_r(r) \simeq 1 - \frac{\alpha^2}{2} \frac{\ell}{r}$

Substituting into (7.6) yields

$$v^2 = \frac{\alpha^2 c^2 \ell}{r} \tag{7.7}$$

7.5 Rigorous Derivation of the Energy-Level Spectrum

Substitute (7.7) into the angular-momentum quantization (7.5): $m_e\sqrt{\frac{\alpha^2 c^2 \ell}{r}}\, rn\hbar$

Simplifying gives

$$r_n = \frac{n^2 \ell}{\alpha^2} \tag{7.8}$$

This is the NO5 version of the Bohr-radius spectrum.

In NO5, orbital energy is a readout of the settlement budget: $E = m_e c^2(\gamma - 1) \simeq \frac{1}{2} m_e v^2$

Using (7.7) and (7.8), we obtain

$$E_n - \frac{m_e c^2 \alpha^2}{2n^2} \tag{7.9}$$

Thus the energy spectrum is uniquely determined by the closure integer n.

7.6 Elliptical Orbits and the Laplace–Runge–Lenz Vector

For a general orbit $r = r(\theta)$, the closure condition (7.2) remains unchanged, while the radial settlement term introduces angular dependence. The orbit equation can be written as

$$\frac{1}{r(\theta)} \frac{1}{p} (1 + e \cos\theta) \tag{7.10}$$

Here:

- $p = n^2 \ell / \alpha^2$;
- e: eccentricity.

In NO5, an ellipse is not a consequence of forces, but the natural solution obtained when the closure integral separates into radial and angular parts. Conservation of the Laplace–Runge–Lenz vector corresponds to the degeneracy of the closure condition with respect to the choice of initial phase.

7.7 Spectral Lines and Transitions: Differences of Closure Integers

A transition is not an "electron jump," but a switch from one set of stable orbits labeled by closure integer n to another set labeled by n'. The released propagation-state energy is

$$\hbar\omega E_{n'} - E_n \frac{m_e c^2 \alpha^2}{2} \left(\frac{1}{n^2} - \frac{1}{n'^2} \right) \tag{7.11}$$

Thus spectral frequencies are completely determined by differences of closure integers.

7.8 Perturbations, Fine Structure, and Closure Breaking

When the closure condition is perturbed (relativistic corrections, magnetic-moment coupling, etc.), $\Phi_s[C] = 2\pi n + \delta\Phi$ the allowed stable solutions split, producing $\Delta E \sim \alpha^4 m_e c^2$

In NO5, fine structure and the Zeeman effect correspond to weak breaking of closure-phase symmetry.

7.9 Isomorphism Between Atomic and Planetary Orbits

Replace ℓ by the macroscopic stepping-closure scale and $\beta_r(r)$ by the Sun's settling-anchor field; equations (7.3)–(7.6) remain valid in the same form. Therefore, atomic and planetary orbits are manifestations of the same closure mathematics at different scales.

7.10 Chapter Conclusions

1. Stable orbits are uniquely selected by the stepping-phase closure integral condition;
2. angular-momentum quantization is the direct algebraic result of the closure integer;
3. the $1/n^2$ atomic spectrum arises from the combination of closure scale and settling anchor;
4. elliptical orbits are natural solutions of the closure condition in general geometry;
5. spectral transitions are switches of closure integers;
6. atomic and celestial orbits are isomorphic in NO5.

Quantization is not a mysterious rule; it is an integer spectrum of closure geometry.

Chapter 8 Transitions, Radiation, Electromagnetic Waves, and Photons: Propagation Solutions of Settling-Loop Failure

Goal of This Chapter

This chapter answers four questions and presents an auditable mathematical structure:

1. What is a transition (not a "particle jump")?
2. Why must a transition radiate (without appealing to empirical rules)?
3. What is an electromagnetic wave (not a causal chain of mutual **E/B** excitation)?
4. What is a photon (not a fundamental particle, but the minimal propagation–settlement packet)?

8.1 The Two-State Boundary: Enclosure State vs. Propagation State (The Only Boundary of This Chapter)

Recalling the previous chapter:

- Enclosure state: the settling phase ϕ_r can complete a local closed loop → existence is possible.
- Propagation state: the settling loop fails → phase is forced into light-speed relay dominated by the stepping phase.

In terms of budget fractions:

- Enclosure state: β_r effective, $\beta_s < 1$.
- Propagation state: $\beta_r \to 0$, $\beta_s \to 1$.

8.2 Ontological Definition of a Transition: Switching the Closure Integer

Chapter 7 established that the core condition for orbits and energy levels is the integer spectrum of stepping-phase closure: $\Phi_s[C] = 2\pi n$

Definition 8.1 (Transition).

A transition is the process by which an enclosure state switches from a stable closure family with

integer n to another stable closure family with integer n'. A transition is therefore not "an electron jumping from point A to point B," but:

- a rewriting of stepping-closure geometry;
- a reconstruction of the settling closed loop;
- a replacement of the internal set of closure paths.

8.3 Why a Transition Must Radiate: Theorem of Residual Expulsion

During a transition, the system must change the closure condition from n to n'within finite time. Since the closure integer changes, a residual necessarily appears: $\delta\Phi \equiv \Phi_s - 2\pi n' \neq 0$

This residual must be handled inside the enclosure state. However, the closure capacity of the settling phase is finite: there exists a maximum sustainable "closure throughput" $\Omega_r^{\max}$. If the phase-rewrite demand generated by the transition exceeds this throughput,

$$\boxed{\dot{\Phi}_{demand} > \Omega_r^{\max}} \tag{8.1}$$

then the enclosure state cannot complete local closure recovery.

Theorem 8.1 (Residual-expulsion theorem).

When an enclosure state cannot reclaim its closure residual during a transition within the throughput of the settling phase, the residual must enter the propagation state and be carried away by the SU network at light speed. This expulsion is radiation.

This requires no "mutual (E)–(B) excitation," only:

- closure must hold;
- settlement throughput is finite;
- overflow implies expulsion.

8.4 Ontological Equation of Electromagnetic Waves: Light-Speed Relay of the Stepping Phase

The propagation state has a single ontology: a phase pattern is relayed through the SU network at speed c. Let $\hat{\mathbf{n}}$ be the unit propagation direction. The scalar phase $\phi(\mathbf{r}, t)$ satisfies the transport equation:

$$\boxed{\partial_t \phi + c\,\hat{\mathbf{n}} \cdot \nabla\phi = 0} \tag{8.2}$$

This is the NO5 ontological propagation equation. It states that an electromagnetic wave is not "field oscillation," but "relay translation of a phase pattern."

8.5 Hard-Core Derivation of Electromagnetic Waves: Discrete SU Lattice, Signs, and the Wave Equation

We now provide the requested hard-core derivation: from the discrete SU lattice constraint "closure residuals cannot accumulate," we derive:

- $\nabla \cdot \boldsymbol{B} = 0$;
- the minus sign in $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$;
- $\nabla \times \mathbf{B} = \frac{1}{c^2}\partial_t \mathbf{E}$;
- the wave equation and propagation speed c.

8.5.1 Discrete setting: edge/face/volume readouts

Discretize space into an SU lattice with spacing ℓ_0 and time step

$$\Delta t = \frac{\ell_0}{c} \tag{8.3}$$

The propagation state allows only "phase updates by relay from neighboring SUs."

- Electric-field readout (edge quantity): discrete gradient of phase

$$E_x(i,j,k)^n \equiv -\frac{\phi^n_{i+1,j,k} - \phi^n_{i,j,k}}{\ell_0} \quad (other\ components\ similarly) \tag{8.4}$$

- Magnetic-field readout (face quantity): oriented "closure-flux" of the minimal closed face.

For example, B_z is attached to the xy-face, with orientation fixed by the right-hand boundary ordering. Key point: **B** is not "excited by **E**" ontologically; it is a face readout of closure orientation.

8.5.2 Discrete curl locked by right-handedness

On an xy-face, define the edge circulation by right-hand ordering (counterclockwise when viewed along +z):

$$(\nabla_h \times \mathbf{E})_z \frac{E_y(i+1,j) - E_y(i,j)}{\ell_0} \frac{E_x(i,j+1) - E_x(i,j)}{\ell_0} \tag{8.5}$$

Note: the sign of each term is fixed by the boundary order, not by convention.

8.5.3 Non-accumulation of closure residuals ⇒ discrete Faraday form (origin of the minus sign)

Propagation requires that the "closure residual" on any minimal face cannot accumulate per tick, otherwise an illegal suspended residue appears.

Therefore, the edge circulation on a minimal face must equal the time update of the face closure flux, with a sign fixed by right-hand closure consistency:

$$(\nabla_h \times \mathbf{E})^n - \frac{\mathbf{B}^{n+1} - \mathbf{B}^n}{\Delta t} \tag{8.6}$$

Ontological origin of the minus sign: the right-hand boundary ordering defines the positive circulation. If the update of **B** had the same sign, it would leave same-sign residuals on the face each tick and accumulate, violating the "non-accumulation" constraint. The only legal pairing is the opposite sign.

8.5.4 No magnetic monopoles: difference identity implies $\nabla_h \cdot \mathbf{B} = 0$

Discrete vector-calculus identity:

$$\nabla_h \cdot (\nabla_h \times \cdot) = 0 \tag{8.7}$$

Taking discrete divergence of (8.6) yields: $\nabla_h \cdot \mathbf{B}^{n+1} = \nabla_h \cdot \mathbf{B}^n$

If initially $\nabla_h \cdot \mathbf{B}^0 = 0$ (no "open-ended" closure faces), then for all n:

$$\boxed{\nabla_h \cdot \mathbf{B}^n = 0} \tag{8.8}$$

This is the hardware proof of "no magnetic monopoles": face readout of closure + difference identity.

8.5.5 Relay continuity ⇒ discrete Ampère–Maxwell form (vacuum)

In the propagation state, closure orientation must also advance with relay update. Symmetrically:

$$(\nabla_h \times \mathbf{B})^n \frac{1}{c^2} \frac{\mathbf{E}^{n+1} - \mathbf{E}^n}{\Delta t} \tag{8.9}$$

This is not "**B** generates **E**," but that changes in closure orientation must match relay updates of the phase-gradient pattern; otherwise non-accumulating residuals would be violated.

8.5.6 Wave equation and speed c: discrete elimination

Taking discrete curl of (8.6) and using (8.9) to eliminate **B** gives:

$$\frac{\mathbf{E}^{n+1} - 2\mathbf{E}^n + \mathbf{E}^{n-1}}{\Delta t^2} c^2 \, \Delta_h \mathbf{E}^n \tag{8.10}$$

Similarly,

$$\frac{\mathbf{B}^{n+1} - 2\mathbf{B}^n + \mathbf{B}^{n-1}}{\Delta t^2} c^2 \, \Delta_h \mathbf{B}^n \tag{8.11}$$

The propagation speed is uniquely fixed by the hardware step size (8.3) to be c. This completes the required "hard-core derivation": signs, minus sign, closure, no monopoles, and wave equations all follow from SU lattice legality—there is no "mutual excitation causal chain."

8.6 Continuous Limit: Maxwell Forms as Readout-Level Closure Constraints

Taking $\ell_0, \Delta t \to 0$, (8.6), (8.8), and (8.9) yield the continuous forms:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{B} = \frac{1}{c^2} \partial_t \mathbf{E} \tag{8.12}$$

Emphasis: these are not ontological propagation causality, but readout-level closure constraints that (8.2) must satisfy in three-dimensional readouts.

8.7 Photon: The Minimal Indivisible Propagation–Settlement Packet

In NO5, propagation ontology is phase relay; "quantization" occurs at settlement endpoints.

Definition 8.2 (Photon).

A photon is the minimal indivisible "record packet of settlement failure" (a minimal phase-residual packet) expelled in a propagation event. Ontologically it is not a particle, but the minimal propagation–settlement unit.

Its energy readout is given by the transition residual:

$$\boxed{E_\gamma = \hbar\omega = E_{n'} - E_n} \tag{8.13}$$

The "quantum" originates from:

- enclosure states allow only integer closure;
- switching closure integers packages residuals in minimal steps for expulsion;
- settlement endpoints account only in minimal packets.

8.8 Universality of Radiation: Ontological Reason Why Acceleration Radiates

Acceleration means a change in the migration rate of phase occupation, increasing closure demand. Whenever $\dot{\Phi}_{demand} > \Omega_r^{\max}$, the enclosure state cannot reclaim residuals and must expel them into a propagation state.

Therefore:

- uniform motion: closure demand is constant and can be internally canceled → no necessary

radiation;

- acceleration: closure demand changes abruptly and exceeds throughput → radiation is necessary.

This is the NO5 ontological explanation of “acceleration radiation.”

8.9 Chapter Conclusions

1. A transition is a switch of the integer closure spectrum, not a position jump;
2. necessary radiation follows from the residual-expulsion theorem;
3. the ontology of electromagnetic waves is the light-speed relay transport equation of the stepping phase;
4. Maxwell forms in the propagation state are readout-level closure legality constraints;
5. the discrete SU lattice derivation locks the minus sign, no monopoles, and the wave equation;
6. a photon is the minimal indivisible propagation–settlement packet, not a fundamental particle.

Electromagnetic waves are not “field mutual excitation,” but the only legal entrustment mechanism by which the SU network carries phase residuals when an enclosure state loses its settling closed loop.

Chapter 9: Hardcore Dilemma of Quantum Mechanics and NO5's Dimensional Reduction Solution: Propagation State—Settlement State, Delayed Accounting, and Bell Violation

Chapter Goal

This chapter does not aim to "reinvent quantum mechanics formulas" but addresses two fundamental tasks:

1. To unify major quantum mechanics dilemmas under the same ontological boundary:
- Propagation state (light-speed relay of stepping phase)
- Settlement state (settling phase loop completion, clicks occur)
2. To replicate the strongest experimental mathematical evidence without invoking “non-local action,” “consciousness collapse,” or “wave function as an entity”:
- The source of interference probability $P \propto |\Psi|^2$.
- The hardware-mutual exclusion of uncertainty $\Delta x \Delta p \geq \hbar/2$.
- The full experimental chain from preparation → transmission → detection → calculation of entanglement.
- Why S can reach $2\sqrt{2}$, without leading to controllable superluminal communication.

9.1 Outline: Propagation State—Settlement State as the Root of Quantum Paradoxes

9.1.1 Definition of Two States (Ontological, Unique Boundary)

- Propagation state: The settling loop fails, $\beta_r \to 0, \beta_s \to 1$, phase patterns are relayed at light speed through the SU network (derived in Chapter 8).
- Settlement state: The settling loop completes, an irreversible “click/spot/record” occurs.

Core judgment: Quantum phenomena are not "the same object being both wave and particle," but the conflation of propagation state and settlement state as one object. This judgment will be mathematically explained in each subsection.

9.2 Interference: Not "Particles Interfering with Themselves," but "Unaccounted Path Ledger"

9.2.1 Multi-path Refresh Amplitude (Readout Level)

Let the set of feasible refresh sequences from source S to screen point x be $\mathcal{P}_{S\to x}$. Define the path weight amplitude:

$$\Psi(x) = \sum_{p\in\mathcal{P}_{S\to x}} w\,(p)\, e^{i\Phi(p)} \tag{9.1}$$

- $w(p) \geq 0$: Achievable weight (determined by boundary conditions/loss/filtering).
- $\Phi(p)$: Stepping phase accumulation along the path (propagation state phase ledger).

9.2.2 Settlement Probability (Click Statistics)

The screen is a high-enclosure structure, which enforces settlement. The statistical frequency satisfies:

$$P(x) \propto |\Psi(x)|^2 \tag{9.2}$$

This is not treating the wave function as an entity, but considering it as the "sum of path ledger weights."

9.2.3 Double Slit: Interference Terms as Stripes

$$\Psi = \Psi_1 + \Psi_2 \Rightarrow |\Psi|^2 = |\Psi_1|^2 + |\Psi_2|^2 + 2R(\Psi_1\Psi_2) \tag{9.3}$$

The stripes originate from the interference term; "one electron at a time still creates stripes" comes from: each time only one settlement occurs, but the statistical preference of the path ledger accumulates over time.

9.3 Measurement Problem: Collapse = Forced Settlement, No Need for "Consciousness Hypothesis"

In NO5:

- Measurement is not an observation act, but pushing the propagation state into the strong-enclosure boundary, forcing the settlement loop to occur.
- The "multi-path ledger" of the propagation state is forced to account for a single result at the moment of settlement.

One line cap:

$$\text{Measurement} \equiv \text{Forced Settlement Event (Loop Completion)} \tag{9.4}$$

Therefore, collapse is not a mysterious process, but a physical event where the loop is completed.

9.4 Uncertainty: Hardware Mutual Exclusion of Two Phase Budgets

In Chapters 1 and 2:

- Stepping phase fraction β_s: Migration readout (momentum side).
- Settlement phase fraction β_r: Locking readout (position side).

Conservation of refresh budget:

$$\beta_s^2 + \beta_r^2 = 1 \tag{9.5}$$

In the same refresh tick, both readouts cannot be optimized simultaneously. Treating "readout precision" as monotonically related to the corresponding budget (this is the minimal assumption for hardware readouts), we get the mutually exclusive lower bound:

$$\Delta x\, \Delta p \geq \frac{\hbar}{2} \tag{9.6}$$

NO5's key point: Uncertainty is not "the world throwing dice," but that the same hardware budget cannot simultaneously perform two mutually exclusive registrations.

9.5 Entanglement: Full Process from Laboratory to Irrefutable CHSH Math Chain

We now focus on the key points: the preparation, transmission, detection, and Bell violation of entanglement.

9.5.1 How Entanglement is Prepared in the Laboratory (SPDC Standard)

(A) Pump source: Provides a highly coherent propagation state ledger.

- Narrow linewidth laser (CW or pulsed) is adopted
- Ontological implication: Supplies a propagation state phase ledger Λ with ultra-low phase noise

(B) Nonlinear crystal: Realizes “legitimate splitting” without triggering settlement.

In a $\chi^{(2)}$ crystal undergoing SPDC, energy conservation and phase matching hold:

$$\omega_p = \omega_s + \omega_i, \quad \mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i \tag{9.7}$$

NO5's audit standard: It's not “one photon becoming two,” but the same propagation state ledger being split into two spatial interfaces, provided that the splitting point does not trigger settlement.

(C) Construction of Entangled States (Taking Polarization Bell States as an Example)

Obtained via Type-II phase matching / double crystals / interference compensation:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A|H\rangle_B + |V\rangle_A|V\rangle_B) \tag{9.8}$$

(D) Filtering and Compensation: Sole Objective – Delaying Settlement

- Narrowband filtering: Extends the coherence time and reduces path information leakage
- Spatial filtering (single-mode fiber): Suppresses premature settlement caused by multimode propagation
- Temperature control/phase compensation: Maintains the sharing of the identical ledger structure by the two interfaces

NO5 Unified Verdict: The common objective of all engineering details is not to enhance entanglement, but to sustain the unsettled state.

9.5.2 How It Propagates: Sending Two Interfaces of the Same Ledger to Two Places

Common links:

- Optical fiber: Loss, dispersion, polarization drift (all induce early settlement).
- Free space/satellite: Background noise, pointing jitter, time synchronization (all equivalently lead to early settlement).

NO5's hard condition is:

$$\tau_{\text{early settlement}} > \tau_{\text{reach both ends}} \tag{9.9}$$

That is: As long as no strong absorption/scattering/thermal noise "readout" occurs before reaching A and B, the ledger remains unaccounted.

9.5.3 How It Detects: Clicks, Time Stamps, and Coincidence Counting

(A) Single-photon detection and timestamps:

The detector (APD or SNSPD) provides click times t_A, t_B.

(B) Coincidence counting: Filters “same ledger events.”

Define the coincidence window Δt, retaining:

$$\boxed{|t_A - t_B| < \Delta t} \tag{9.10}$$

NO5's key point here: Coincidence counting is not a statistical trick, but a filtering operator for "same unaccounted ledger." Without this step, ideal Bell violation values would not be obtained.

9.6 Hardcore Math: From Bell State to $S = 2\sqrt{2}$

9.6.1 Measurement Model: Polarization Analyzer and ±1 Output

Let the measurement direction at endpoint A be a, at endpoint B be b. The measurement results are ± 1.

For the state $|\Phi^+\rangle$, quantum mechanics gives the correlation function (standard result):

$$E(a, b) = \cos 2\,(a - b) \tag{9.11}$$

Here, the "2" arises from the double angle relationship of polarization under rotation.

NO5's explanation:

$|\Phi^+\rangle$ represents the "shared orientation structure" of the same ledger at two interfaces. The measurement angle is simply a rotation of the interface readout base.

9.6.2 CHSH Combination and Maximum Value

CHSH Definition:

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b') \tag{9.12}$$

Using the classical optimal angles:

$$a = 0, \quad a' = \frac{\pi}{4}, \quad b = \frac{\pi}{8}, \quad b' = -\frac{\pi}{8} \tag{9.13}$$

Substitute into (9.11):

$$E(a, b) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$$

$$E(a, b') = \cos(\pi/4) = \frac{\sqrt{2}}{2}$$

$$E(a', b) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$$

$$E(a', b') = \cos(3\pi/4) = -\frac{\sqrt{2}}{2}$$

Thus:

$$S = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right) = 2\sqrt{2} \tag{9.14}$$

This step is pure algebra and irrefutable: As long as the correlation function is $\cos 2\,(a - b)$, the maximum violation is $2\sqrt{2}$.

9.6.3 Where the "Irrefutable Point" Lies: Failure of Bell's Factorization Assumption

Bell's key premise for the (local hidden variable) derivation is probability factorization:

$$P(A, B|\lambda) = P(A|\lambda)\,P(B|\lambda) \tag{9.15}$$

NO5's core judgment: In the unaccounted ledger state Λ, this factorization does not hold because A and B are not independent objects but two interfaces of the same ledger.
In the shortest mathematical language:
Λ unaccounted $\Rightarrow$ Independence factorization fails $\Rightarrow |S| > 2$ No need for non-local action (9.16)

9.7 Why Entanglement Cannot Be Used for Superluminal Communication (Mathematical Closure: No Signal)

Even with strong correlations, the marginal distribution on one endpoint must not depend on the settings of the other endpoint. For the Bell state, the marginal probability at endpoint A is:

$$P(A = +1|a) = \sum_{B=\pm 1} P\,(A = +1, B|a, b) \tag{9.17}$$

For $|\Phi^+\rangle$, we can calculate:

$$P(A = \pm 1|a) = \frac{1}{2} independent of b \tag{9.18}$$

Therefore, communication is not possible because:
The change at the other endpoint only alters the joint distribution, not the local marginal distribution.
NO5's explanation:
An unaccounted ledger only allows "reading," not "writing." Any attempt to write will trigger early settlement, which disrupts entanglement.

9.8 The Strict Meaning of "Pseudo-Entanglement" (Avoid Misreading)

NO5's "pseudo-entanglement" does not negate the experiment but negates the modern narrative of "two independent particles being connected."
Experimental correlation is real; the object-level narrative of "long-distance connection" is pseudo.
Mathematically, this means: Factorization in (9.15) cannot be used to describe the unaccounted state Λ.

9.9 Unified Solution to the Core Dilemmas of Quantum Mechanics (Final Synthesis)

1. Wave-particle duality: Propagation state vs. settlement state, no longer contradictory.
2. Measurement/collapse: Forced settlement event, no need for consciousness hypothesis.
3. Uncertainty: Hardware mutual exclusion lower bound between two phase budgets.
4. Entanglement and Bell violation: Shared ledger multi-interface readings, failure of independence factorization.
5. No superluminal communication: Marginal distribution does not depend on the other endpoint settings (no signal), closed by (9.18).

Top-level conclusion: Quantum mechanics' "strangeness" is not natural but stems from conflating propagation and settlement states as the same object, leading to ontological confusion. NO5 uses two-phase refresh budget and "unaccounted ledger" interface structure to unify all paradoxes as different manifestations of the same closure principle.

9.10 Chapter Summary

This chapter rigorously replicates experimental statistics (interference, $CHSH = 2\sqrt{2}$, no signal)

while providing an ontological explanation that does not rely on non-local action:
Entanglement is the multi-interface reading of an unaccounted propagation ledger, and Bell's factorization premise fails in this state.

Chapter 10 Newton's Three Laws, Solar Rotation, and Solar-System Orbits: The Macroscopic Limit of Settlement Consistency and Stepping Closure

Goal of This Chapter

This chapter proves that macroscopic mechanics is not the "ontology of forces," but the selection outcome of motions that can be closed. Newton's three laws, stellar rotation, and planetary orbits are unified by two master lines:

- Master line A (readouts): $\beta_s^2 + \beta_r^2 = 1, \quad v = \beta_s c, \quad d\tau = \beta_r dt$
- Master line B (closure):

$$\boxed{\oint_C \omega_s \, dt = 2\pi n} (n \in \mathbb{Z}) \tag{10.1}$$

10.1 The Macroscopic Entrance: "No Suspended Phase"

Macroscopic enclosure states do not allow propagating suspended phase. The linkage has not disappeared; it has sunk into a settlement-consistency constraint. Therefore macroscopic dynamics reduces to two kinds of variables:

- stepping phase (migration readout);
- settling phase (readouts of existence/time/mass).

10.2 The Settling-Fraction Field and a "Refraction-Type" Law of Motion

Define the settling-fraction field (anchor field):

$$\boxed{\beth(\mathbf{r}) \equiv \beta_r(\mathbf{r})} \tag{10.2}$$

Its readout gradient gives "gravity":

$$\boxed{\mathbf{g}(\mathbf{r}) = -c^2 \nabla \ln \beth(\mathbf{r})} \tag{10.3}$$

For any trajectory, the normal curvature satisfies:

$$\boxed{v^2 \kappa = |\mathbf{g}_\perp|} \tag{10.4}$$

Acceleration is not caused by a force; the settlement gradient requires curvature.

Part I Ontological Rewriting of Newton's Three Laws

10.3 The First Law (Inertia)

Theorem 10.1.

If, in a given $\beth(\mathbf{r})$, stepping closure has no residual over the macroscopic time window, then the trajectory requires no correction.

Inertia = closure without deficit.

10.4 The Second Law (Acceleration)

Multiply the curvature law (10.4) by the enclosure-scale readout m:

$$ma_{\perp} = m|\mathbf{g}_{\perp}|$$

"Force" is merely the readout that prices the required curvature by the enclosure scale.

10.5 The Third Law (Action–Reaction)

Theorem 10.2.

Settlement consistency of the SU network requires conservation of total correction; any local correction must be accompanied by an equal and opposite compensation.

This is not mutual pushing; it is that the ledger cannot be settled unilaterally.

Part II Solar Rotation: Necessity of Internal Closure

10.6 The Internal-Closure Theorem

$$\boxed{\omega\,\ell = c} \tag{10.5}$$

Theorem 10.3.

For an enclosure body whose interior contains sustained stepping flow without leakage, the only stable realization is axial closure; since relay speed is c, one must have $\omega\ell = c$.

10.6.1 Why the Sun Must Rotate

If a star did not rotate, its internal stepping flow could not close stably and would leak into a propagation state, destroying its long-term stability as a settling anchor. Rotation is not an initial condition; it is a geometric necessity of internal closure.

Part III Solar-System Orbits: Necessity of External Closure

10.7 External Closure Condition

For a planet to move stably in the Sun's ($\beth(r)$, it must satisfy:

$$\frac{v^2}{r} = c^2 \frac{d}{dr} \ln \beth\,(r) \tag{10.6}$$

and simultaneously satisfy the integer closure condition (10.1).

10.8 Weak-Field Limit → Newtonian Gravity and Kepler's Laws

Take the weakest outer deviation: $\beth(r) = 1 - \epsilon(r), \quad \epsilon \ll 1$

Choose the minimal long-range symmetric form: $\epsilon(r) = \frac{GM}{rc^2}$

Substituting back yields:

$$\boxed{g(r) = \frac{GM}{r^2}} \tag{10.7}$$

Hence:

$$\boxed{v^2 = \frac{GM}{r}}, \quad \boxed{T^2 = \frac{4\pi^2}{GM} r^3} \tag{10.8}$$

The Newton–Kepler laws emerge naturally.

10.9 Elliptical Orbits and Precession

The general solution satisfies the Binet form. When $\beth(r)$ contains higher-order terms, closure

acquires a small residual, and the system repairs the ledger by rotating the closure direction:

$$\boxed{\Delta\varpi \simeq 2\pi\kappa} \tag{10.9}$$

Part IV Unifying Internal and External Closure

10.10 Unified Form

- external linear readout: $v = \beta_s c$
- internal circumferential closure: $\omega\ell = c$

Both arise from the same fact: stepping relay speed is fixed at c. The only difference is where closure occurs (internal vs. external).

10.11 Division of Labor

- solar rotation: internal closure (volume return of stepping flow);
- planetary orbits: external closure (closable paths in $\beth(r)$).

10.12 Chapter Conclusions

1. Newton's three laws are the macroscopic limit of "closure without residue";
2. solar rotation follows from the internal-closure theorem $\omega\ell = c$;
3. solar-system orbits are determined by the gradient of the anchor field $\beth(r)$;
4. Newtonian gravity and Kepler's laws appear naturally in the weak-field limit;
5. atomic and planetary orbits are isomorphic, both being integer spectra of stepping closure.

The macroscopic universe is not driven by "forces," but selected by which motions can close.

Chapter 11 Ontological Rewriting of Relativity: Light-Speed Refresh, Orthogonal Phases, and Readout Invariance

Goal of This Chapter

Relativity has long been interpreted as "geometric deformation of spacetime." In the NO5 framework, this narrative is reduced to a deeper fact: invariance of the speed of light does not come from spacetime symmetry, but from the hardware invariance of the space-unit (SU) refresh protocol; relativistic effects are not stretching or compression of spacetime, but changes in readouts produced by redistributing the refresh budget between the stepping and settling phases. This chapter proves that:

- all special-relativistic effects follow from the readout consequences of $\beta_s^2 + \beta_r^2 = 1$;
- the equivalence principle of general relativity follows from the spatial distribution of $\beta_r(\mathbf{r})$;
- the Lorentz transformation is a readout transformation that preserves the refresh budget, not a coordinate "trick."

11.1 Root Explanation of Light-Speed Invariance (Not an Assumption)

Axiom recall (Chapter 1): the refresh-relay speed of each SU is identically c. This implies:

- whenever refresh is fully transferred into the stepping phase, its linear readout must be c;
- no refresh relay can be "faster" or "slower."

Hence light-speed invariance is not an empirical input, but:

$$light \equiv propagation\ state\ due\ to\ settling - loop\ failure \Rightarrow\ v = c \tag{11.1}$$

It is independent of the source and the observer.

11.2 The Unique Structural Variable of Relative Motion: Refresh Fractions

Let the relative motion between two observers be described by the stepping-phase fraction β_s. Then:

$$v = \beta_s c \tag{11.2}$$

Refresh-budget conservation gives:

$$\beta_s^2 + \beta_r^2 = 1 \tag{11.3}$$

This is the master identity of the chapter.

11.3 Time Dilation: Reduced Density of Settlement Events

In NO5, time is not a parameter but a count of settlement events. Define:

$$d\tau = \beta_r \, dt \tag{11.4}$$

Using (11.3) and (11.2), $\beta_r = \sqrt{1 - \frac{v^2}{c^2}}$

hence:

$$d\tau = dt \sqrt{1 - \frac{v^2}{c^2}} \tag{11.5}$$

Transparent interpretation: it is not that "time slows," but that refresh budget for settlement is occupied by stepping, so fewer settlement events can be completed per unit external time.

11.4 Length Contraction: Insufficient Registration of Spatial Addresses

In NO5, the readout of length is defined as the number of consecutive space units whose settlement locking can be completed per unit time. When an object moves:

- the stepping phase occupies refresh budget;
- the settlement budget fraction β_r used for spatial-address locking decreases;
- thus fewer spatial addresses can be registered per unit time.

Mathematically:

$$L = L_0 \, \beta_r = L_0 \sqrt{1 - \frac{v^2}{c^2}} \tag{11.6}$$

Emphasis: the ontology of space is not compressed; what changes is the registration efficiency.

11.5 Unified Origin of the Lorentz Factor

Define the Lorentz factor: $\gamma \equiv \frac{1}{\beta_r}$

then:

$$\gamma \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \tag{11.7}$$

In NO5:

- γ is not a dynamical amplification factor;
- it is an amplified readout of settlement budget relative to external time.

11.6 Mass–Energy Relation: Energy Pricing of Settlement Budget

Let the settlement-scale readout of an enclosure state at rest be m_0. Define energy as the pricing of settlement budget:

$$E = \gamma m_0 c^2 \tag{11.8}$$

When $\beta_s = 0$,

$$E_0 = m_0 c^2 \tag{11.9}$$

Transparent interpretation: mass–energy equivalence is not "mass turning into energy," but the settlement budget required to maintain existence being priced in energy units.

11.7 Momentum: Linear Readout of the Stepping Phase

Momentum is not "something carried by an object," but the linear-direction refresh readout of the stepping phase. Thus:

$$\mathbf{p} = \gamma m_0 \mathbf{v} \tag{11.10}$$

This is naturally compatible with the energy–momentum four-vector relation, but requires no a priori Minkowski geometry.

11.8 Ontological Rewriting of the Lorentz Transformation

11.8.1 The Traditional Issue

Traditionally, Lorentz transformations are obtained by requiring "light speed invariance," but do not explain why such invariance must hold.

11.8.2 NO5 Transformation Principle

Principle: any legitimate coordinate transformation must preserve the SU refresh budget—i.e., preserve the form of (11.3).

For one-dimensional relative motion, require:

$$c^2 dt^2 - dx^2 = c^2 d\tau^2 = c^2 \beta_r^2 dt^2 \tag{11.11}$$

This yields directly:

$$dt'\& = \gamma \left(dt - \frac{v}{c^2} dx \right), dx'\& = \gamma (dx - v\, dt) \tag{11.12}$$

Interpretation: the Lorentz transformation is not a "spacetime postulate," but a readout transformation that keeps the refresh-budget decomposition invariant.

11.9 Equivalence Principle: Spatial Distribution of the Settlement Fraction

In Chapter 10 we introduced the settlement-fraction field $\beta_r(\mathbf{r})$. The time readout generalizes to:

$$d\tau(\mathbf{r}) = \beta_r(\mathbf{r})\, dt \tag{11.13}$$

When β_r varies in space:

- settlement rhythm differs by location;

- in the readout layer this appears as “gravitational time dilation.”

The local acceleration readout is:

$$\mathbf{g}(\mathbf{r}) = -c^2 \nabla \ln \beta_r\,(\mathbf{r}) \tag{11.14}$$

Ontological form of the equivalence principle: an accelerated frame and a gradient of the settlement fraction are equivalent because both are, in essence, changes in the spatial distribution of β_r.

11.10 Unified Readout Interpretation of General-Relativistic Effects

- gravitational redshift: differences in settlement rhythm due to different β_r;
- free fall: sliding along paths that are “easier to settle”;
- orbital bending: the minimal-curvature solution for stepping closure in the field $\beta_r(\mathbf{r})$.

NO5 requires no ontological assumption of “spacetime curvature”; all these effects are geometric consequences of settlement-fraction readouts.

11.11 Applicability Boundary of the Continuous Limit and Relativity

Relativistic descriptions are valid when:

observation scale $\gg \ell_0, \quad \beta_r \nrightarrow 0$]

When:

- $\beta_r \to 0$ (black-hole horizon), or
- observation approaches ℓ_0 (SU scale),

continuous relativistic descriptions fail and one must return to discrete refresh and closure criteria.

11.12 Chapter Conclusions

1. light-speed invariance arises from the SU refresh protocol, not from spacetime assumptions;
2. relativistic effects arise from refresh-budget redistribution between stepping and settling phases;
3. time dilation, length contraction, and mass–energy relations are readout effects;
4. Lorentz transformations preserve the refresh budget;
5. the equivalence principle follows from the spatial distribution of $\beta_r(\mathbf{r})$;
6. general-relativistic effects are geometric manifestations of settlement-fraction gradients.

In NO5, relativity is not about “how spacetime changes,” but about “how refresh is budgeted.”

Chapter 12 Superconductivity: Limit Workpoint of Settling Loops and Macro Manifestation of Phase Consistency

Goal of the Chapter

This chapter proves a core proposition: superconductivity is not determined by "electron pairing" in material details, but naturally emerges as a limit working state when the settling phase achieves nearly perfect closure on the macroscopic scale.

Under the NO5 framework, this chapter will provide a unified explanation of:

1. The ontological origin of zero resistance;
2. Why the Meissner effect inevitably occurs;
3. Why the Josephson effect naturally has phase and frequency calibration;
4. Why superconductivity is the reality anchor of “engineered settling civilization.”

12.1 Rearticulation of the Superconductivity Problem (Why Traditional Explanations Are Insufficient)

Traditional theories (e.g., BCS) explain superconductivity from material microdetails, which is successful in engineering but leaves three unresolved ontological questions:

- Why is "phase" so fundamental in superconductivity, yet lacks a clear physical definition?
- Why does the superconducting state repel, rather than shield, external magnetic fields?
- Why does superconductivity provide cross-scale phase consistency (macroscopic quantum phenomenon)?

NO5's approach is to start not from materials but from the refresh–closure structure, tracing back to the ontological conditions for superconductivity.

12.2 Ontological Criterion for Superconductivity: Near-Perfect Settling Loop Closure

Reviewing the core structure from Chapter 4:

- Current = directional migration of phase occupation (stepping phase);
- Resistance = dissipation caused by repetitive failure of small closures in materials due to geometric inconsistencies;
- Heat = statistical manifestation of failed closures.

Definition 12.1 (Superconducting State)

When the material structure enables the settling phase to form a highly consistent, repeatable, low-residual closure path across the entire macroscopic channel, and the settling loop hardly ever fails, the system enters the superconducting state.

In budget language:

$$\Omega_r^{loss} \approx 0,\ \beta_r^{eff}\ \ remains\ consistent\ across\ the\ channel \tag{12.1}$$

12.3 Zero Resistance: Not "No Scattering," but "No Need for Re-Settlement"

12.3.1 The NO5 Ontology of Resistance

In Chapter 4, resistance was strictly defined as: dissipation caused by the failure of the settling phase to achieve consistency in the material due to geometric inconsistencies.

12.3.2 Why Superconductivity Must Have Zero Resistance

When the superconducting state forms:

- The settling phase completes closure across the entire loop in one pass;
- There is no need for repetitive small closures in local regions;
- Thus, there are no persistent failure events.

Mathematically, this means:

$$R \propto \frac{\text{Settlement Failure Rate}}{\text{Phase Mobility}} \to 0 \tag{12.2}$$

Conclusion: Zero resistance is not an "idealization" of the material but the direct result of achieving maximal consistency in the settling structure.

12.4 The Meissner Effect: Geometric Exclusion of External Settling

12.4.1 The NO5 Ontology of Magnetic Fields

In NO5, a magnetic field is defined as a directional closure structure formed by the settling phase being forced outward during continuous migration of the stepping phase.

12.4.2 Why Superconductivity Repels Magnetic Fields

Inside a superconductor:

- The settling loop has already been perfectly completed;
- There is no "external settling" requirement for the system;
- Any external attempt to introduce an external settling structure would disturb internal closure consistency.

Thus, the system's only stable response is:

$$\text{Expelling external settling (magnetic flux) from the system} \tag{12.3}$$

This is not shielding, but the geometric exclusivity of closure consistency.

12.5 The Josephson Effect: Synchronized Readout of the Settling Phase Across Junctions

12.5.1 Continuity of the Settling Phase

In superconductors, the settling phase ϕ_r has a continuous definition at macroscopic scales:

$$\phi_r(\mathbf{r}) \in \mathbb{R}$$

When two superconductors are connected via a thin insulating layer:

- The stepping phase is suppressed;
- The settling phase can still form cross-junction consistency constraints.

12.5.2 DC Josephson Effect

A phase difference $\Delta\phi_r$ produces a steady-state current:

$$I = I_c \sin(\Delta\phi_r) \tag{12.4}$$

NO5 explanation: This is a periodic condition of achievable settlement closure across junctions.

12.5.3 AC Josephson Effect

When there is voltage (V) (settling potential difference) across the junction:

$$\frac{d}{dt}\Delta\phi_r = \frac{2e}{\hbar} V \tag{12.5}$$

This gives the frequency–voltage relation:

$$f = \frac{2e}{h} V \tag{12.6}$$

Key Point: This is not an empirical coincidence but a direct manifestation of the settling phase as a refresh counter, thus becoming the most stable voltage–frequency calibration in nature.

12.6 Superconductivity and Consistency of Relativistic Readout

In Chapter 11, the time readout is defined as: $d\tau = \beta_r\, dt$

An important feature of superconductivity is:

- β_r remains highly consistent across the macroscopic loop;
- Therefore, different positions' settling rhythms naturally synchronize.

This explains:

- Macroscopic quantum coherence;
- Long-term phase stability in superconducting loops;
- The Josephson junction becoming a "relativistic readout consistency device."

12.7 Superconductivity as the Reality Anchor of "Settling Civilization"

Returning this chapter to the full structure, we can see superconductivity's unique position:

- In electromagnetic engineering: → zero-dissipation limit (pinnacle of Chapter 4's

engineering);

- In information: → macroscopic phase consistency (real starting point for field communication);
- In dynamics: → controllable settling throughput (necessary prerequisite for mass reduction and propulsion).

Thus, superconductivity is not a niche topic, but the first bridge from NO5 theory's "explaining nature" to "engineering civilization."

12.8 Superconductivity and Black Hole Limit Symmetry (Deep Structural Connection)

In Chapter 11, a black hole is characterized by:

- $\beta_r \to 0$: Settling depletion
- $\ell \to \ell_0$: Saturation of closure scale

The superconducting state is its mirror image:

- β_r is maximized and consistent across the channel;
- Settling closure is nearly perfect at the macroscopic scale.

Both form the two end limits of NO5's "limit structure picture":

Extreme state	Settling structure
Black hole	Settling depletion, outward propagation
Superconductivity	Perfect settling, no external usage

This provides a symmetrical closure for NO5's "extreme structural picture."

12.9 Final Conclusions of the Chapter

1. Superconductivity is the extreme working state of near-perfect settling loop closure;
2. Zero resistance arises from the settling failure rate tending to zero, not scattering disappearance;
3. The Meissner effect is the geometric exclusivity of closure consistency;
4. The Josephson effect is the inevitable manifestation of synchronized settling phase across junctions;
5. Superconductivity provides a real physical anchor for field communication, continuous computing, and space propulsion;
6. Superconductivity and black holes form the two extreme limits of the settling structure.

In NO5, superconductivity is not an anomaly but the upper limit of "how well settling can be achieved."

Chapter 13 Black Holes and the Density-Consistency Theorem: A Dual-Criterion Limit of Settlement Depletion and Stepping-Closure Saturation

Overall Goal of This Chapter

This chapter reduces black holes from "geometric singularities" to a NO5 limit state:
black hole = a limit enclosure state in which settlement phase is depleted ($\beta_r \to 0$) and stepping closure cannot be further compressed ($\ell \to \ell_0$) simultaneously.

From this definition, the chapter derives directly:

1. what the event horizon is (an ontological boundary, not geometric mysticism);
2. why singularities do not exist;

3. why “density consistency” not only holds but is strengthened into a non-retreatable theorem;
4. why black-hole thermodynamics follows an area law;
5. falsifiable observational differences implied by the framework.

13.1 NO5 Ontological Variables for Black Holes: $\boldsymbol{\beta_r}(\mathbf{r})$ and $\boldsymbol{\ell}(\mathbf{r})$

Recall from Chapter 11:

- settlement fraction field (anchor field):

$$\boxed{\beth(\mathbf{r}) \equiv \beta_r(\mathbf{r})} \tag{13.1}$$

- gravity readout (settlement gradient):

$$\boxed{\mathbf{g}(\mathbf{r}) = -c^2 \nabla \ln \beth(\mathbf{r})} \tag{13.2}$$

This chapter introduces a second ontological variable: the stepping-closure scale $\ell(\mathbf{r})$, which satisfies the established internal-closure relation:

$$\boxed{\omega(\mathbf{r})\,\ell(\mathbf{r}) = c} \tag{13.3}$$

and has a lower bound imposed by SU discreteness:

$$\boxed{\ell(\mathbf{r}) \geq \ell_0} \tag{13.4}$$

Equation (13.4) is the key final closure for “density consistency.”

13.2 Dual-Criterion Definition of the Black-Hole Limit (Stronger than the Standard View)

Definition 13.1 (Black-hole limit state).

An enclosure state enters the black-hole limit in a region $\mathcal{V}$ if:

1. settlement depletion

$$\boxed{\beth(\mathbf{r}) \to 0 \quad (\mathbf{r} \in \mathcal{V})} \tag{13.5}$$

2. stepping closure reaches the minimal-scale limit

$$\boxed{\ell(\mathbf{r}) \to \ell_0 \quad (\mathbf{r} \in \mathcal{V})} \tag{13.6}$$

In this case, the region enters the black hole limit.

The first condition locks “vanishing time readout,” while the second locks “density cannot continue to increase.”

13.3 Event Horizon: The Settlement Boundary ($\beth = 0$) as the Geometric Boundary of “Closure Impossibility”

Theorem 13.1 (Horizon condition).

The event horizon $\mathcal{H}$ is defined by

$$\boxed{\beth(\mathbf{r}) = 0, \quad \mathbf{r} \in \mathcal{H}} \tag{13.7}$$

Its physical meaning is:

- the settlement-phase budget is zero;
- no enclosure-settlement event can occur within that surface;
- an external observer can never obtain new settlement records from the interior.

This is the ontological origin of "information cannot escape," without invoking coordinate singularities.

Equivalent stepping statement (more geometric).

From (13.3)–(13.4):

- if closure would require $\ell < \ell_0$, closure is impossible;
- the horizon is the boundary of the "closure-impossible region."

Thus the horizon can also be characterized by:

$$\omega_{req}(\ell) \equiv \frac{c}{\ell} has no solution on \ell \geq \ell_0 \tag{13.8}$$

The settlement criterion and the closure criterion merge at the horizon.

13.4 Time Freezing and Redshift: Settlement Failure, Not Geometric Divergence

From the ontological relation in Chapter 2:

$$d\tau = \beth\, dt \tag{13.9}$$

As ($\beth \to 0$),

$$\boxed{d\tau \to 0} \tag{13.10}$$

This is not "weird time coordinates," but the disappearance of settlement events: time readout as settlement counting naturally fails. Redshift likewise follows from "settlement rhythm cannot be maintained," with no need to treat "spacetime stretching" as ontology.

Part I No Singularities and the Density-Consistency Theorem (Core)

13.5 Why Singularities Do Not Exist: Two Hard Limits Seal the Route

Traditional singularities arise from "unbounded continued compression." In NO5, compression must satisfy two conditions simultaneously:

1. settlement fraction continues to decrease ($\beth \to 0$);
2. stepping-closure scale continues to shrink ($\ell \to 0$).

But NO5 imposes two hard limits:

- $\beth \geq 0$, and 0 is already the extreme (it cannot be "more negative");
- $\ell \geq \ell_0$ (it cannot be smaller than the SU minimal scale).

Therefore any attempt to drive density $\rho \to \infty$ must violate (13.4). A singularity is not merely "unlikely"; it is physically impossible—there is no legal path.

13.6 Density-Consistency Theorem: A Strict Proof by Double Locking

To emphasize the strengthened, non-retreatable nature, we give an explicit proof structure.

Definition: enclosure density and closure scale.

Interpret "enclosure density" as the number density of minimal units that can maintain closure per unit volume. In NO5, this density scales inversely with the minimal closure scale:

$$\rho_{enc} \propto \frac{1}{\ell^3} \tag{13.11}$$

This is geometric counting: smaller ℓ allows more closure units per volume.
Step 1: compression is equivalent to shrinking ℓ.
Further "inward compression" necessarily requires ℓ to decrease.
Step 2: closure angular frequency must rise.
From (13.3),

$$\omega = \frac{c}{\ell} \tag{13.12}$$

Step 3: when ℓ reaches the lower bound ℓ_0, ω reaches an upper bound.
At $\ell = \ell_0$,

$$\omega_{max} = \frac{c}{\ell_0} \tag{13.13}$$

Step 4: further compression would require illegal conditions (no solution).
Further compression implies $\ell < \ell_0$ or $\omega > \omega_{\max}$, both violating the NO5 hardware constraint. Hence ρ_{enc} cannot be increased beyond its maximum.
Theorem 13.2 (Density-consistency theorem).

$$\ell \geq \ell_0 \quad \Rightarrow \quad \rho_{enc} \leq \rho_{\max} \propto \frac{1}{\ell_0^3} \tag{13.14}$$

Once the interior enters the limit enclosure state, it necessarily reaches and maintains this bound:

$$\rho_{BH}(\mathbf{r}) = \rho_{\max(\mathbf{r} \in \mathcal{V})} \tag{13.15}$$

This is "density consistency": there is no higher-density center and thus no singularity.
What "strengthened density consistency" means: it is no longer a heuristic, but a strict geometric upper-bound theorem enforced by the lower bound ℓ_0.

13.7 Growth Mechanism: Not Central Collapse, but Outward Extension of the Saturation Region

When external matter crosses the horizon:
it is forced toward $\beth \to 0$;
and it expands the saturated region where $\ell \to \ell_0$.
Thus black-hole growth is not "falling into a singularity," but:

$$R_{\mathcal{H}} \text{ moves outward so that more region enters } \beth = 0, \ell = \ell_0 \tag{13.16}$$

The interior maintains density consistency; only the radius of the saturated region extends outward.

Part II NO5 Derivation of Black-Hole Thermodynamics (Ontological Origin of the Area Law)

13.8 Horizon Temperature: Minimal Fluctuation Scale of the Settlement Gradient

In NO5, temperature ontologically means "density of settlement events / fluctuations of settlement failure." At the horizon, $\beth$ drops sharply from a positive value to 0, producing a strong gradient:

$$|\nabla \beth|_{\mathcal{H}} \tag{13.17}$$

Propagation-state fluctuations are expelled from the near-horizon region; their characteristic energy scale is set by the "minimal divisible settlement packet":

$$k_B T_H \sim \hbar \frac{c}{L_{grad}}, \quad L_{grad} \sim \frac{\beth}{|\nabla \beth|} \tag{13.18}$$

In the ($\beth \to 0$) boundary layer, L_{grad} is of order R_s, yielding $T_H \propto \frac{1}{M}$, consistent with the Hawking-temperature scaling.

Key point: temperature comes from fluctuations of the settlement boundary layer, not from a "virtual particle" narrative.

13.9 Black-Hole Entropy: Horizon Interface Counting ⇒ Area Law

In NO5, "information/entropy" is not an internal microstate count, but the number of independent settlement interfaces that can be formed. The horizon is a settlement boundary, and the minimal interface area is set by ℓ_0. Hence the interface count is:

$$N_{iface} \sim \frac{A}{\ell_0^2} \tag{13.19}$$

Entropy is proportional to the interface count:

$$S \propto N_{iface} \propto \frac{A}{\ell_0^2} \tag{13.20}$$

This is the NO5 ontological origin of the area law: not statistical mysticism, but boundary interface counting.

Part III Information and Falsifiable Predictions

13.10 NO5 Reconstruction of the Information Problem

In NO5, "information" means writable settlement records. Inside a black hole ($\beth = 0$):

- new settlement records cannot be produced;
- it is not that information vanishes, but that the system "stops being writable."

Hawking radiation corresponds to propagation-state fluctuations expelled from the horizon boundary layer and does not require leakage of interior information.

13.11 Falsifiable Prediction Points

NO5 black holes may be equivalent to GR in the external macroscopic region, but differ in the following discriminants:

1. No-singularity prediction: the interior is a density-consistent saturated region, implying potentially different post-merger echoes or late-time tail modes.
2. Boundary-layer fluctuation spectrum: higher-order corrections to temperature and radiation spectrum depend on the detailed shape of ($|\nabla \beth|$).
3. Post-merger relaxation spectrum (QNMs): if the interior is not a GR singularity structure, some mode frequencies and damping rates should exhibit systematic shifts.
4. Upper limit for extremal spin: the new stepping-closure limit from $\omega\ell = c + \ell \geq \ell_0$ may yield testable differences near the Kerr extremal regime.

13.12 Chapter Conclusions

1. A black hole is defined by dual criteria: $\beth \to 0$ (settlement depletion) and $\ell \to \ell_0$ (closure scale at its limit);

2. the event horizon is a settlement boundary and a closure-impossibility boundary;
3. time freezing is settlement failure, not a coordinate artifact;
4. singularities are sealed off by two hard constraints: settlement cannot be "more depleted," and closure scale cannot be smaller than ℓ_0;
5. the density-consistency theorem is strengthened into a strict upper-bound theorem: $\rho_{BH} = \rho_{\max} \propto 1/\ell_0^3$
6. black-hole thermodynamics arises from horizon interface counting and boundary-layer fluctuations, yielding the area law naturally;
7. multiple falsifiable observational pathways are provided.

A black hole is not a spacetime catastrophe, but a stable limit state reached when the SU refresh structure satisfies settlement depletion + closure-scale saturation.

Chapter 14: Field Communication and Continuous Computers: Synchronization of Settlement Phases and the Ontological Rewriting of Computation

Chapter Goal

This chapter demonstrates that communication and computation are not secondary technical issues, but inevitable manifestations of the refresh-settlement structure in the information domain. The core proposition of this chapter is: when information is no longer understood as "bits carried by propagation states," but as "structural changes in the realizability of settlement phases," the physical paradigms of communication and computation will undergo a fundamental shift.

14.1 From "Signal Propagation" to "Settlement Feasibility": Ontological Rewriting of Information

14.1.1 Traditional Information Theory's Physical Assumptions

Traditional communication and computation assume:

1. Information is carried by discrete symbols (bits);
2. Information must propagate through some medium (electromagnetic waves, electron flow);
3. Communication delays and bandwidth are determined by the propagation process.

In NO5, this equates to: information is forcibly bound to the propagation state of the stepping-phase. This binding is not inevitable, but a historical choice.

14.1.2 Information Definition in NO5

Definition 14.1 (Information)

Information is not a symbol or energy but the structural changes in which settlement states are achievable and which are not.

The change in information is not about "what has been sent," but rather about the change in the set of realizable settlement states. This concept is the common foundation for "field communication" and "continuous computation."

14.2 Field Communication: Synchronization of Settlement Phases, Not Propagation

14.2.1 Ontological Definition of Field Communication

Definition 14.2 (Field Communication)

Field communication refers to: two or more systems synchronizing their realizable settlement states through shared or associated settlement-phase structures, without relying on the propagation

state to transfer information bit-by-bit.

Key points:

- Synchronization ≠ Propagation
- What changes is the "feasible settlement structure," not the "carrying medium."

14.2.2 Physical Feasibility of Settlement Synchronization

In NO5, the settlement phase has the following properties:

1. It does not carry the concept of propagation speed;
2. It is only read out when settlement events occur;
3. If multiple systems share the same settlement constraints (or projections), their settlement states are not independent.

This is structurally similar to “entanglement = multi-interface readout of the unbalanced propagation ledger” from Chapter 9, but the operating points differ:

Entanglement	Field Communication
Microscopic	Macroscopic
Propagation-state dominated	Settlement-state dominated
Incontrollable	Engineer-controllable
Cannot be written	Adjustable

14.2.3 Minimal Mathematical Expression of Field Communication

Let the settlement phase readings for systems A and B be $\phi_r^{(A)}, \phi_r^{(B)}$.

The synchronization constraint for field communication can be written as:

$$\phi_r^{(A)}(t) - \phi_r^{(B)}(t) \in \mathcal{C} \tag{14.1}$$

where $\mathcal{C}$ is the set of allowed phase-difference values determined by the system's structure. Communication does not mean "writing bits," but rather: changing the structure of the set $\mathcal{C}$.

14.3 Why Field Communication Does Not Violate Causality or the Speed of Light Limit

A natural question arises: if the propagation state is not involved, would causality be violated? NO5 provides a rigorous negative answer.

14.3.1 Settlement Phases Cannot Be "Written"

Settlement phases can only be read when settlement occurs, and cannot be arbitrarily manipulated to specified values.

Therefore:

- Cannot “specify” the settlement result for B at endpoint A;
- Can only change which results are allowed.

Mathematically, $P\ (result_B | structure_A) \neq \delta$ (specified value), which is completely consistent with “no-signaling theorem” in Chapter 13.

14.3.2 True Boundary of Causality

In NO5, causality is not determined by propagation speed, but by the rule: any change that can be specified must propagate via the stepping phase; non-propagating synchronizations can only alter feasibility, not specify results. Field communication strictly adheres to this rule.

14.4 Continuous Computers: Ontological Shift in Computation

14.4.1 Physical Limits of Discrete Computing

Traditional computers define computation as:

- Discrete states
- Discrete time steps
- Discrete logic gates

In NO5, this corresponds to: forcibly discretizing settlement phases into “0/1” and using stepping-phase propagation to drive flipping.

This introduces three bottlenecks:

- Energy consumption grows exponentially with frequency;
- Delay is dominated by interconnect propagation;
- Parallelism depends on stacking rather than on the underlying structure.

14.4.2 Ontological Definition of Continuous Computers

Definition 14.3 (Continuous Computers)

A continuous computer is a system where the continuous structure of the settlement phase is the computational variable, and “minimizing closure residuals” serves as the system's computational dynamics.

In this paradigm:

- Computational state = settlement-phase distribution $\phi_r(\mathbf{r})$;
- Computational evolution = continuous rearrangement of closure residuals;
- Computational final state = stable closure structure (attractor).

14.5 Minimal Dynamics Equation for Continuous Computing

Let the settlement-phase field be $\phi_r(\mathbf{r}, t)$, and define the closure residual functional:

$$\mathcal{F}[\phi_r] = \int d^3 r[(\nabla\phi_r)^2 + V(\phi_r)] \tag{14.2}$$

The natural evolution of continuous computing is gradient flow:

$$\partial_t \phi_r - \frac{\delta\mathcal{F}}{\delta\phi_r} \tag{14.3}$$

This is not an “algorithm” set by humans, but the fastest residual elimination needed for settlement-phase closure.

14.6 Why "Computation Results" Must Be Stable

Under this dynamics:

- Any initial state will evolve along $-\nabla\mathcal{F}$;
- The final state satisfies:

$$\frac{\delta\mathcal{F}}{\delta\phi_r} = 0 \tag{14.4}$$

This corresponds to structures with zero or minimal closure residuals. The “correctness” of the computation is not determined by logical decisions, but by physical stability.

14.7 Ontological Advantages of Continuous Computing for Complexity Problems

Many NP-hard problems are essentially:

- Path closure
- Topological consistency
- Conflict minimization

In NO5, these problems are naturally mapped to: $\min_{\phi_r} \mathcal{F}[\phi_r]$

The system evolves directly towards the attractor solution through physical evolution, not by enumerating states.

14.8 Ontological Distinction Between Continuous Computing and Quantum Computing

Quantum Computing	Continuous Computing
Propagation-state interference	Settlement-state rearrangement
Coherence easily destroyed	Stability comes from closure
Output probability	Output structure
Requires error correction	Does not rely on error correction

Continuous computing is not a refinement of quantum computing, but a completely different ontological computing paradigm.

14.9 Unified Structure of Field Communication and Continuous Computing

Both in NO5 are unified as: manipulation of the realizability structure of settlement phases.

- Field Communication: controlling “cross-system feasibility synchronization”;
- Continuous Computing: controlling “internal-system feasibility rearrangement.”

14.10 Final Conclusions of This Chapter

1. Information in NO5 is redefined as the structure of settlement feasibility;
2. Field communication synchronizes realizability through settlement-phase synchronization, not bit-by-bit transmission;
3. Field communication does not violate causality because the settlement result cannot be specified;
4. Continuous computing evolves based on the minimization of closure residuals, not on logic gate flipping;
5. The correctness of computation is ensured by physical stability;
6. Field communication and continuous computing are manifestations of the same physical structure in the communication and computation domains.

When computation and communication are reduced to "how settlement happens," information technology directly engages with physical ontology for the first time.

Chapter 15: Mass Reduction and Space Propulsion: Controllable Failure of Settlement Loops and the Ontological Rewriting of Spatial Anchors

Chapter Goal

In NO5, "mass", "inertia" and "gravity" are not independent entities, but rather readouts of the occupancy and consistency constraints of settlement phases within the SU network. Thus:

- Mass reduction is not "anti-gravity" or "removal of matter", but the reduction of settlement proportion readouts;
- Spatial propulsion is not "pushing via force application", but the rearrangement of closable paths in space, such that an object is "adsorbed" along the new direction of the minimum

closure residual.

This chapter introduces a key conclusion: the only path for mass reduction and propulsion is the controllable rewriting of settlement closure throughput capability and closure geometry.

15.1 The Ultimate Ontology of Mass: "Feasibility Reading" of Settlement Proportions

Recalling Chapters 1–2: ： $\Omega_0^2 = \Omega_s^2 + \Omega_r^2$, $\beta_r = \frac{\Omega_r}{\Omega_0}$, $\beta_s = \frac{\Omega_s}{\Omega_0}$, $\beta_s^2 + \beta_r^2 = 1$

In NO5:

- Speed reading: $v = \beta_s c$
- Time reading: $d\tau = \beta_r dt$
- Energy magnification: $\gamma = 1/\beta_r$

For macroscopic structures, "mass" is not a geometric stack, but the scale of settlement phase allocation required for the structure to maintain existence.

Thus, the effective mass reading is:

$$m_{\text{eff}} \propto \langle \Omega_r \rangle_{\text{structure feasibly achievable}} \equiv \Omega_0 \langle \beta_r \rangle \tag{15.1}$$

This is key:

Mass reduction = lowering $\langle \beta_r \rangle$, not changing material composition.

15.2 The Unique Physical Mechanism of Mass Reduction: Controllable Introduction of Settlement Closure "Throughput Insufficiency"

15.2.1 Vacant Phases and the Strict Boundary of the Macroscopic

We have negated the existence of propagating vacant phases at the macroscopic scale. However, "denying propagation" does not equate to "denying emergence"; what is allowed macroscopically is:

Local, short-term, recoverable failure of settlement loops (insufficient closure throughput).

This does not leak in the form of wave propagation (that is radiation), but can form a reduced settlement feasibility within the structure.

15.2.2 Closure Throughput Criterion (Engineering Mother Formula)

Let the rate of demand for closure in the structure be $\dot{\Phi}_{demand}$, and the maximum closure throughput be $\Omega_r^{\max}$. When:

$$\dot{\Phi}_{demand} > \Omega_r^{\max} \tag{15.2}$$

The system will experience unavoidable "closure deficit."

If prevented from outward expulsion by structural constraints, this manifests as:

$$\beta_{r,\text{eff}} \beta_r - \Delta\beta_r, \Delta\beta_r \propto \frac{\dot{\Phi}_{demand} - \Omega_r^{\max}}{\Omega_0} \tag{15.3}$$

Thus, effective mass decreases:

$$m_{\text{eff}} = m_0 (1 - \kappa\, \Delta\beta_r) \tag{15.4}$$

where κ is the structure valuation coefficient (related to packaging size, geometric channel count).

This is the only legitimate entry for mass reduction: not relying on a stronger static field, but on making closure insufficient.

15.3 Why "Stronger Electric and Magnetic" Does Not Equal Mass Reduction (Hardcore Refutation)

Many intuitively believe that "stronger electromagnetic fields" will generate stronger anomalous effects. The NO5 conclusion is opposite:

- Strong static fields enhance closure capacity (mobilize more available closure paths).
- Static enhancement raises $\Omega_r^{\max}$, making (12.2) harder to satisfy.
- Therefore, it only "reinforces packaging," not reducing mass.

Written in NO5 criteria:

$$\dot{\Phi}_{demand} \uparrow \Omega_r^{\max} \uparrow \text{ Faster.} \quad \text{If simultaneously } \Rightarrow \Delta\beta_r \approx 0 \tag{15.5}$$

Thus, mass reduction is not a "field strength problem," but a "throughput mismatch problem."

15.4 "Mass Reduction" and "Time Reading" Are Co-originating: Why "Relativistic Effects" Appear

From $d\tau = \beta_r dt$, if β_r is lowered:

$$d\tau_{eff} = (\beta_r - \Delta\beta_r)dt \tag{15.6}$$

Thus, mass reduction is always accompanied by a trend of "time reading decrease":

- It is not the change in time itself,
- But a decrease in the density of settlement events.

This is why "mass reduction engineering" is naturally associated with "time potential difference/clock offset"—both are β_r reading changes.

Second Part: Ontological Rewriting of Space Propulsion

15.5 Propulsion is Not "Applying Force," But "Rearranging Closure Paths in Space"

Chapter 10 proved that macroscopic motion is the feasible path of stepping closure in the settlement anchor field. Therefore, to achieve propulsion, it is not about "applying force," but about changing "which path is most feasible to close."

This gives the ontological definition of propulsion:

Propulsion = creating an asymmetrical distribution of β_r feasibility in space, causing the closure minimal residual path to be biased in one direction.

15.6 Minimum Mathematical Criterion for Propulsion: Difference of Forward and Reverse Closure Feasibility

Let the effective settlement proportion in the direction of the flight for the front and rear be:

$$\beta_r^{(+)} \equiv \beta_r(\mathbf{r} + \delta\hat{\mathbf{n}}), \quad \beta_r^{(-)} \equiv \beta_r(\mathbf{r} - \delta\hat{\mathbf{n}})$$

If there exists:

$$\beta_r^{(+)} < \beta_r^{(-)} \tag{15.7}$$

The forward closure is more difficult (closure is more deficient), and the rear closure is easier.

To maintain overall consistency, the system will be "attracted" along the "easier to close" direction (or equivalently, rejected from the direction of more difficult closure).

This point can be written as an equivalent acceleration reading (Chapter 10 gravitational reading form):

$$\mathbf{a}_{eff} - c^2 \nabla \ln \beta_r \|_{eff} \tag{15.8}$$

The essence of propulsion is creating a directional structure of $\nabla \ln \beta_r$, not creating "thrust."

15.7 Propulsion and Electromagnetism: Why Must It Rely on "Controllable Failure of Settlement Loops"

In electromagnetic engineering, the only actively controllable components are:

- Migration of phase occupation (current)
- External magnetic fields (settlement-phase external closure)
- Radiation/propagation states (failure of closure expulsion)

But propulsion does not require radiation itself; rather, it involves creating a controllable closure mismatch inside the packaging structure, thereby directionally changing the β_r distribution.

Thus, the minimum structural requirement for propulsion is:

1. Strong variation rate (increase $\dot{\Phi}_{demand}$)
2. Asymmetry (difference between front and back, establish (12.7))
3. Suppress expulsion (do not let all the deficits turn into radiative escape)
4. Maintain structural stability (avoid entering destructive propagation states)

These four must come together to form the "propulsion window."

15.8 The Ontological Explanation of "Space Falling Propulsion" (Space Falling)

Writing propulsion as "falling" is more accurate in NO5:

- Gravitational/acceleration readings are already $-c^2 \nabla \ln \beta_r$
- When a strong directional β_r gradient is artificially created
- The object "falls" along this gradient is inevitable

Thus:

Space propulsion is not rocket propulsion, but artificially creating settlement anchor point gradients.

Third Part: Engineering Limits and Correspondence with Black Holes (Grand View of the Limits)

15.9 Black Holes as the Opposite Mirror Image of "Limit Mass Reduction States"

Chapter 11 gives the black hole dual criteria:

- $\beta_r \to 0$ (settlement exhaustion)
- $\ell \to \ell_0$ (closure scale saturation)

This indicates:

- Black holes are not "infinite density," but "saturation region after settlement exhaustion to the limit"
- For engineering, it provides a "limit reference":

It is not possible to push β_r to 0 without entering disastrous structural instability

Thus, the reasonable goal of mass reduction engineering is: local, small-scale, recoverable $\Delta\beta_r$, not black hole-like extremes.

15.10 Civilizational Blueprint: From Electromagnetic Civilization to "Settlement Civilization"

At this point, we can present a grand yet rigorous vision:

- Electromagnetic civilization: controlling propagation states (electromagnetic waves)

- Settlement civilization: controlling settlement closure (β_r distribution)
- Propulsion civilization: controlling directional closure paths (engineering of $\nabla \ln \beta_r$)

Among which:

- Field communication and continuous computing (previous chapter) are the information aspects of "settlement civilization"
- Mass reduction and propulsion are the energy aspects of "settlement civilization"
- Black holes are the extreme boundaries of "settlement civilization"

15.11 Final Conclusions of This Chapter

1. Mass is the reading of settlement allocation: $m_{eff} \propto \langle \beta_r \rangle$.
2. The only mechanism of mass reduction is closure throughput mismatch: $\dot{\Phi}_{demand} > \Omega_r^{\max}$ leading to $\Delta\beta_r > 0$
3. Propulsion is not force, but the directional rewriting of β_r distribution: $\mathbf{a}_{eff} = -c^2 \nabla \ln \beta_r$.
4. "Strong fields" do not necessarily work; the key is the rate of change, asymmetry, and suppression of expulsion.
5. Black holes provide a limit reference for engineering: $\beta_r \to 0$ and $\ell \to \ell_0$ saturation boundaries.

Mass reduction and space propulsion in NO5 are not mysterious technologies, but the controllable failure of settlement loops and the engineering rewriting of settlement anchor gradients.

Appendix A

Physical Origin of Calculus: The Continuous Limit of Refresh Differencing

A.1 Motivation (Why This Appendix Is Needed)

Modern physics assumes it may freely use:

- derivatives $\frac{d}{dt}$
- gradients ∇
- limits $\lim_{\Delta} \to 0$

yet rarely answers a more fundamental question: why does nature "allow" these limiting operations? In NO5, this question can be answered in a single stroke.

A.2 Refresh Differencing as the Primitive Object

In NO5, time and space are not continuous manifolds; they are built from the discrete refresh of SUs:

- minimal spatial scale: ℓ_0
- minimal time step: $\Delta t_0 = \frac{\ell_0}{c}$

Therefore, the primitive physical operation is not differentiation, but differencing:

$\Delta f f(t + \Delta t_0) - f(t)$

A.3 Physical Definition of the Derivative (Not a Mathematical Postulate)

Definition A.1 (Physical derivative).

In NO5, a derivative is not an "infinitesimal ratio," but

$$\frac{df}{dt} \equiv \lim_{N\to\infty} \frac{1}{N\Delta t_0} \sum_{k=1}^{N} [f(t + k\Delta t_0) - f(t + (k-1)\Delta t_0) r] \tag{A.1}$$

Its physical meaning is a statistical average of many discrete refresh differences over a macroscopic observation window.

A.4 Why the Limit Exists (Core Conclusion)

The physical reason the limit exists is not "mathematical continuity," but that macroscopically observable quantities must be insensitive to refresh-phase noise. If some quantity is

- highly sensitive to a single refresh,
- then it is macroscopically non-repeatable and non-measurable.

Thus nature "permits" only those quantities for which $\lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} \Delta f_k$ converges to become stable physical readouts.

Conclusion A.1 (Existence of limits theorem, physical form).

Calculus limits hold because physical observables must converge in the refresh statistical limit; quantities that do not converge cannot become stable readouts of nature.

A.5 Direct Implications for Classical Mechanics and Field Theory

- Newtonian acceleration $a = \frac{dv}{dt}$: a statistical readout of stepping differencing;
- electromagnetic field evolution $\partial_t \mathbf{E}$: statistical evolution of phase-gradient differencing;
- quantum phase $\exp(iS/\hbar)$: exponential encoding of accumulated refresh phase.

Appendix B

Physical Origin of Limits and Continuous Symmetries: Why Noether's Theorem Holds

B.1 Where Symmetry Comes From (Not "Mathematical Beauty")

Noether's theorem states:

- time translation → energy conservation;
- space translation → momentum conservation;
- rotation → angular momentum conservation.

But the deeper question is: why does nature "respect" these continuous symmetries?

B.2 Refresh-Budget Conservation as the Unique Source

In NO5, the only strictly conserved quantity is the SU refresh budget:

$$\Omega_0 = \text{refresh budget of each SU}$$

This budget is

- independent of position;
- independent of time;
- independent of direction.

Therefore:

- time translation does not change the refresh ledger;
- space translation does not change the refresh ledger;

- rotation does not change the refresh ledger.

B.3 Physical Rewriting of Noether's Theorem

Theorem B.1 (Noether–NO5 form).

Continuous symmetry is not an abstract property of spacetime, but the invariance of the refresh budget under re-labeling operations in the statistical limit.

Mathematically "continuous symmetry" is physically equivalent to: Ω_0 invariant under translation/rotation.

B.4 Why the Quantum Layer Inherits These Symmetries

Quantum-state phase is $\psi \sim e^{i\Phi}$, where Φ is the accumulated readout of refresh phase.

Therefore:

- translation does not change the accumulation rule;
- rotation does not change the closure condition.

Quantum symmetry is not new; it is the low-level readout of refresh symmetry.

Appendix C

Physical Legitimacy of Continuous Fields and Partial Differential Equations

C.1 Why Partial Differential Equations Can Be Written

Classical field theory assumes: $\partial_t f(\mathbf{r}, t), \quad \nabla f(\mathbf{r}, t)$

In NO5, fields are not ontological continua—so why are PDEs effective?

C.2 Field = Statistical Density of Refresh States

In NO5,

- a field $f(\mathbf{r}, t)$ is not an attribute of a single SU;
- it is the statistical average of SU refresh states within a unit volume.

Thus: $f(\mathbf{r}, t)\langle f_{SU}\rangle_{V(\mathbf{r})}$

C.3 Physical Meaning of Partial Derivatives

Definition C.1 (Physical partial derivatives).

$$\partial_t f = \frac{1}{V}\sum_{SU\in V}\frac{f_{SU}(t+\Delta t_0)-f_{SU}(t)}{\Delta t_0} \tag{C.1}$$

$$\nabla f = \frac{1}{V}\sum_{SU\in V}\frac{f_{SU}(\mathbf{r}+\ell_0)-f_{SU}(\mathbf{r})}{\ell_0} \tag{C.2}$$

This shows that PDEs are the "enclosure language" of SU refresh differencing under a joint space–time statistical limit.

C.4 Why PDEs Can Predict Propagation, Waves, and Steady States

Because:

- refresh is local;
- differencing rules are uniform;
- the statistical limit is stable;

therefore, wave equations, diffusion equations, and Laplace/Poisson equations become repeatable macroscopic laws.

Appendix D

Failure Criterion of the Continuous Limit: When Calculus Stops Working

This appendix is deliberately "hard," and practically useful.

Theorem D.1 (Failure criterion of the continuous-limit approximation).

Calculus approximations fail if and only if: observation scale $\lesssim \max(\ell_0, c/\omega_{\max})$

Physical meaning:

- near the SU scale;
- near the black-hole limit;
- near ultra-high-frequency settlement-failure regions.

In these regimes:

- classical orbits fail;
- field theory fails;
- one must return to discrete SU descriptions.

One-sentence global summary of the appendices

Calculus and limits are not "mathematical gifts from nature," but expressions permitted by the SU refresh structure in the macroscopic statistical limit; NO5 specifies both their validity and their failure boundaries.

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